



The bridge to possible

Cisco SD-Access LISP Solution Fundamentals

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BRKENS-2810

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Cisco Webex App

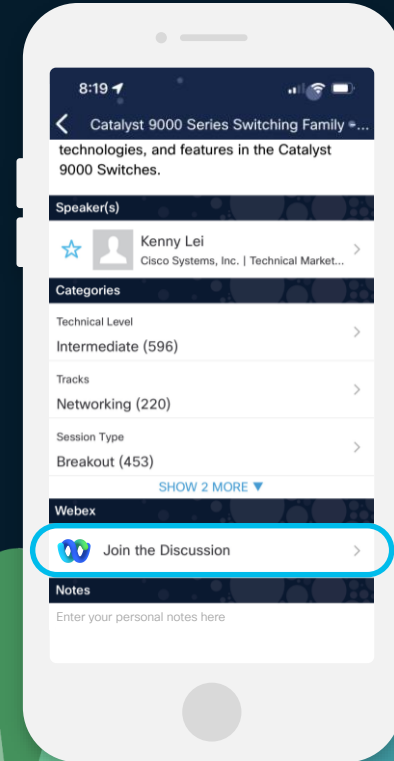
Questions?

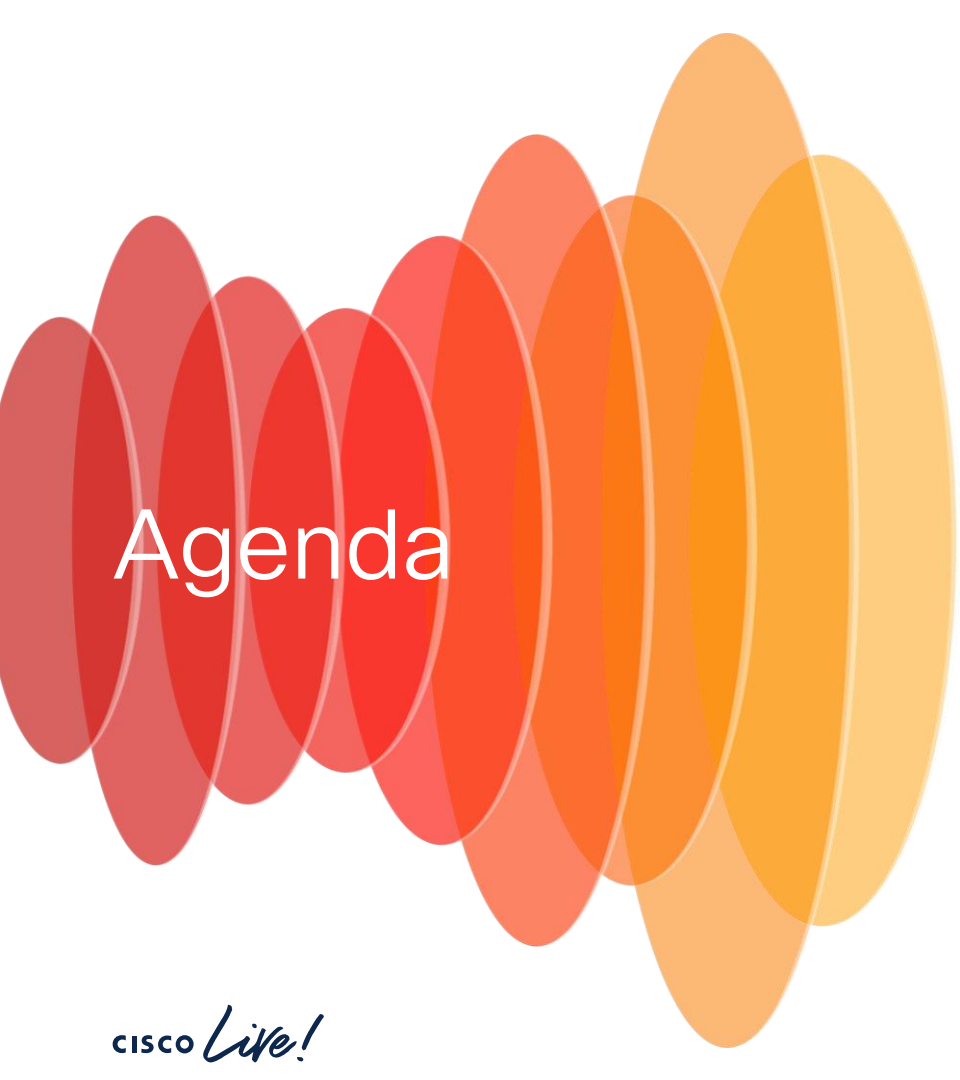
Use Cisco Webex App to chat with the speaker after the session

How

- 1 Find this session in the Cisco Live Mobile App
- 2 Click “Join the Discussion”
- 3 Install the Webex App or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

Webex spaces will be moderated by the speaker until June 7, 2024.

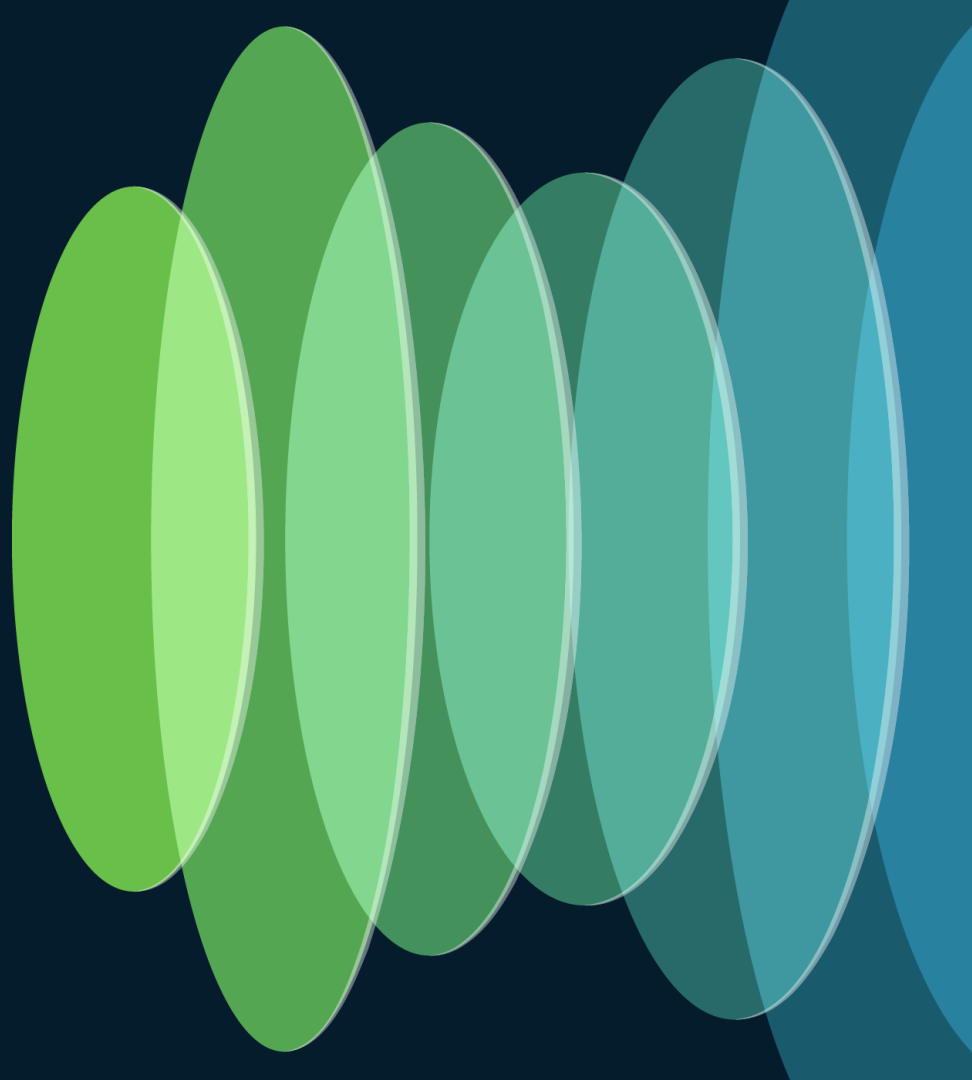




Agenda

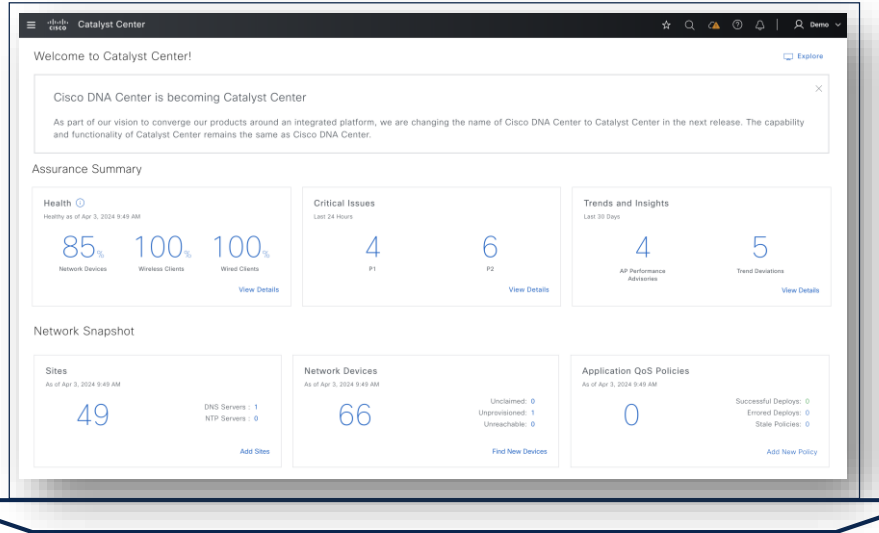
- Why Cisco SD-Access LISP?
- Roles and Terminology
- Fabric Fundamentals
- Multiple Fabric
- Conclusion

Why Cisco SD-Access LISP?



Intent-Driven Automation

Standardized templates for consistent implementation



Easy for admins to configure and manage networks

Configuration workflows to drive intent



Create Layer 3 Virtual Networks

Create a Layer 3 Virtual Network to host interconnect IP subnets.



Create Fabric Site

Create and configure an SD-Access Fabric Site and Fabric Zones. Fabric Zones are optional and reside within Fabric Sites.



Create Anycast Gateways

Create a gateway to attach IP pools to Layer 3 Virtual Networks.



Configure Multicast

Configure multicast routing within Cisco SD-Access Layer 3 Virtual Networks.



Create Extranet Policy

Create an Extranet Policy to share services between a Provider Virtual Network and one or more Subscriber Virtual Networks.

... and many more!

Intent-Driven Automation

Automation with highly flexible scale



Create Anycast Gateways

Create a gateway to attach IP pools to Layer 3 Virtual Networks.



Best practice configurations curated and validated extensively for each workflow.

Simple UI based guided workflow for the administrator to deploy with a few clicks.

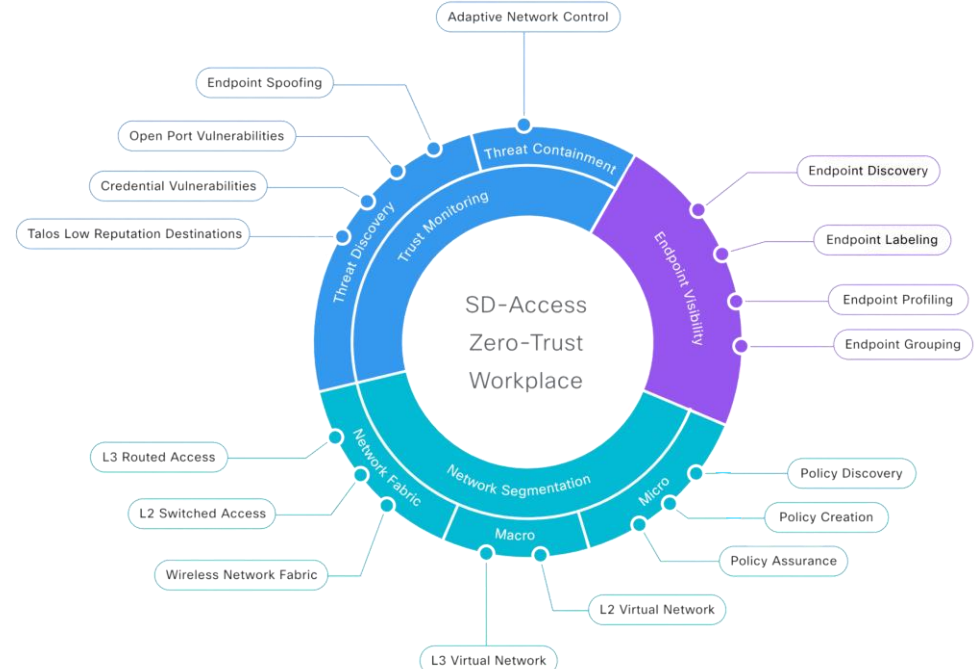
Configure multiple devices across or within a site in one-go!

SD-Access Zero-Trust Workplace

Based on the Organization's goal, explore and achieve Zero Trust for workplace.

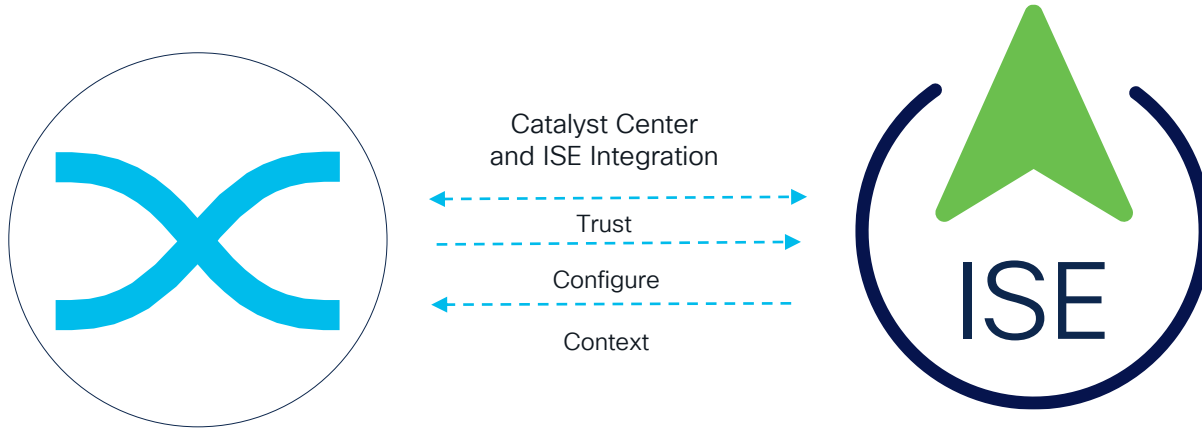


3 Pillars of SD-Access Zero Trust Workplace:
Visibility, Segmentation, and Trust.



Attribute-Based Policy and Segmentation

Catalyst Center and ISE integration facilitates automated configuration and segmentation at scale



Automated configuration and segmentation policies
paired with scalable onboarding of endpoints.



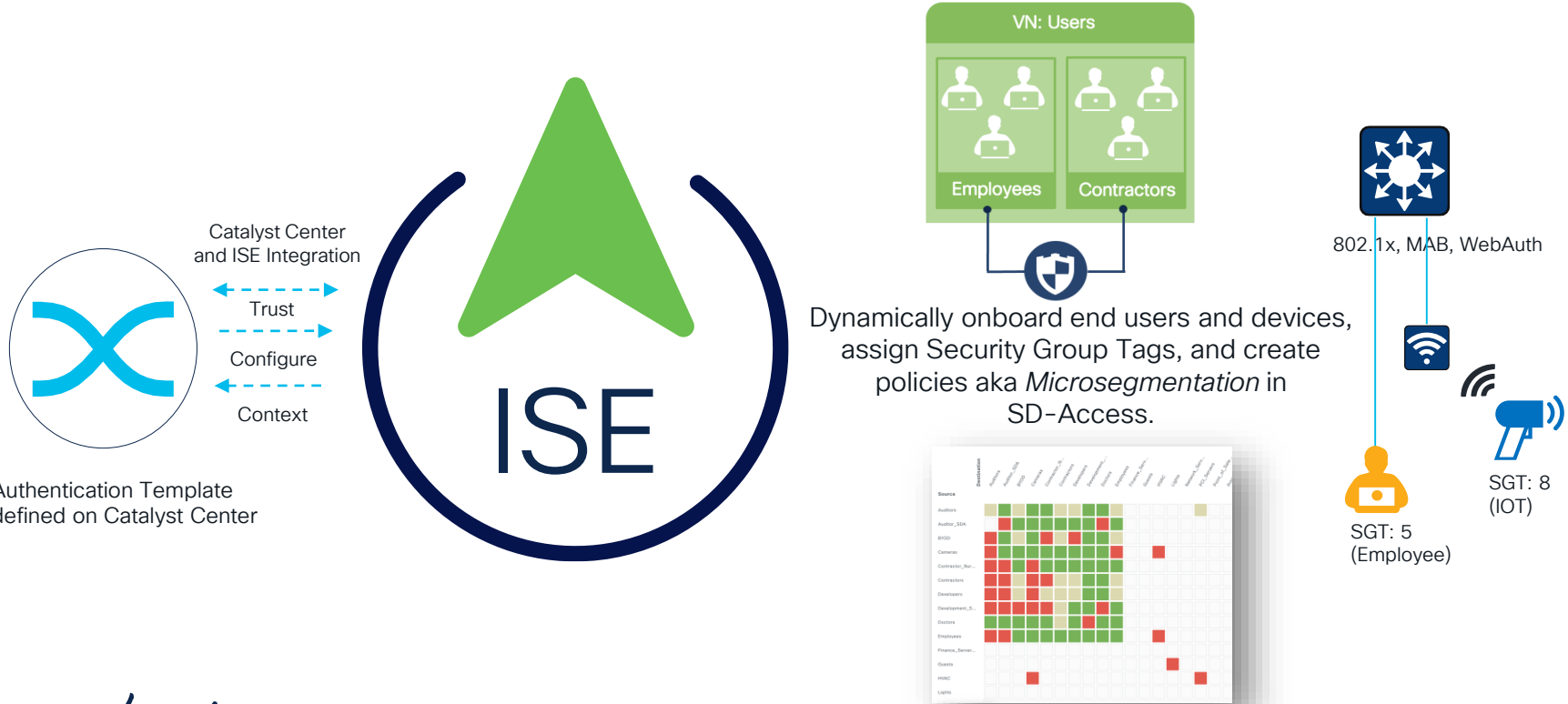
Attribute-Based Policy and Segmentation

Centralize policy definition on ISE for wired and wireless endpoints



Attribute-Based Policy and Segmentation

Move away from IPs and ACLs to Group Based Access Control





Additional
Information

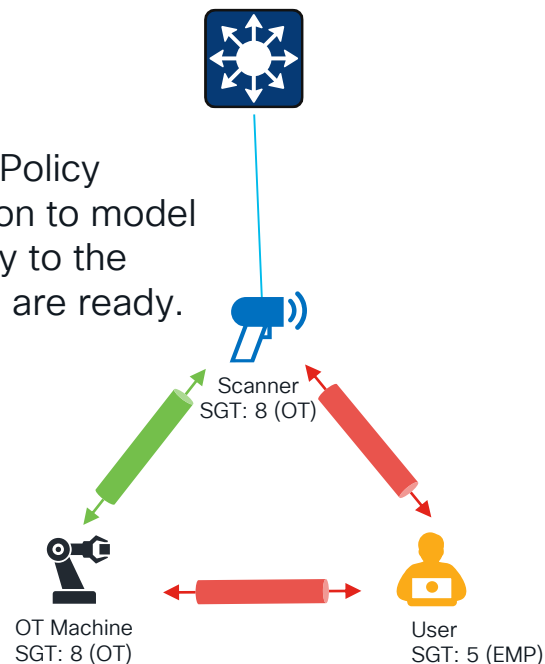
Attribute-Based Policy and Segmentation

Analytics and Visibility with ISE



Group Based Policy Analytics

Use Group Based Policy Analytics application to model policies and deploy to the network when you are ready.



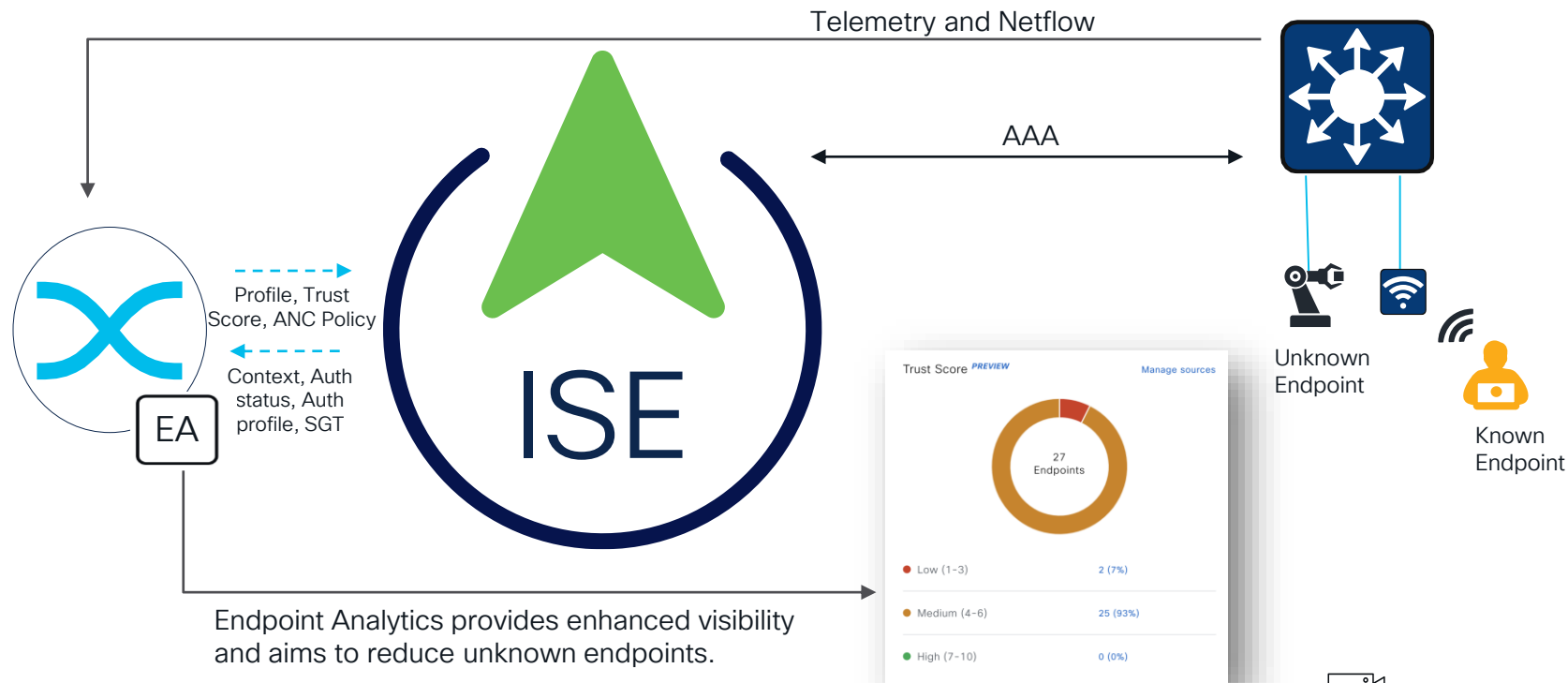
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Additional
Information

Attribute-Based Policy and Segmentation

Analytics and Visibility with ISE



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Tailormade for Campus Networks

Bringing together best of wired and wireless

Campus =
People + Things

University • Large Corporations • Banks
Healthcare • Manufacturing

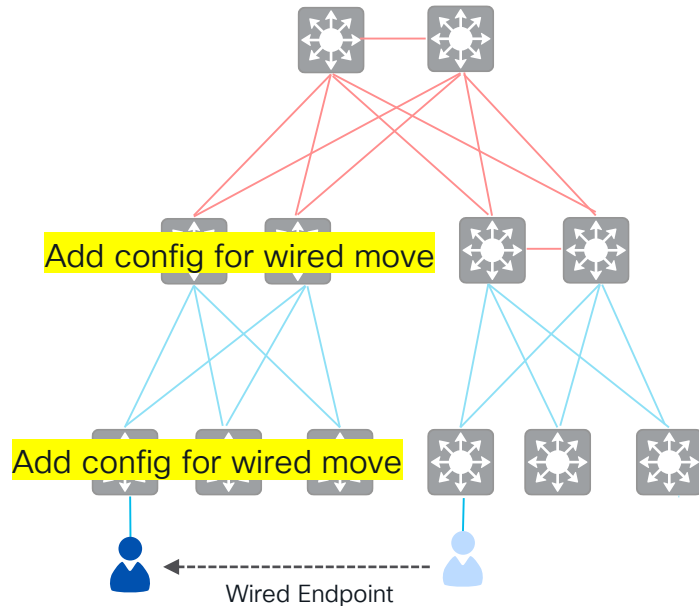
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Tailormade for Campus Networks

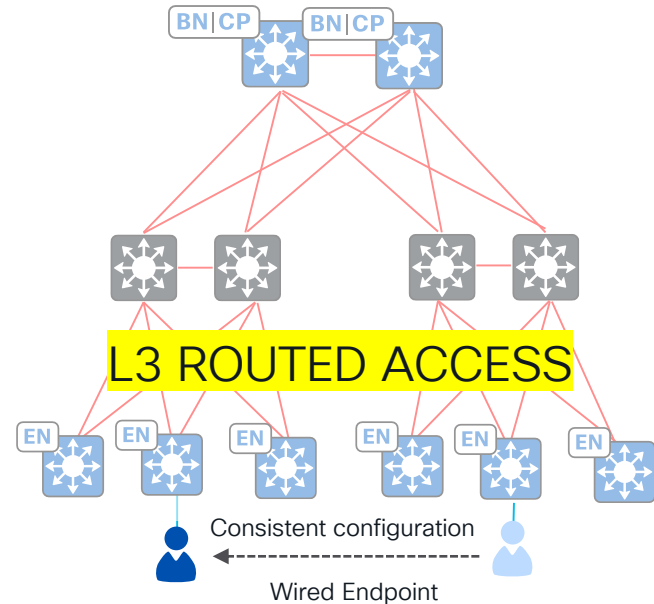
Seamless and secure mobility of endpoints (Wired use-case)

Traditional Network



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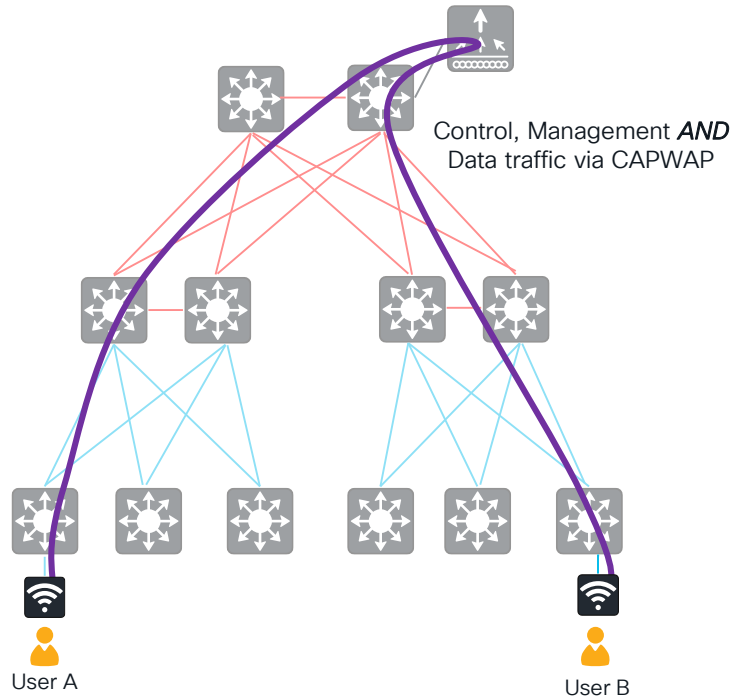
Software-Defined Access



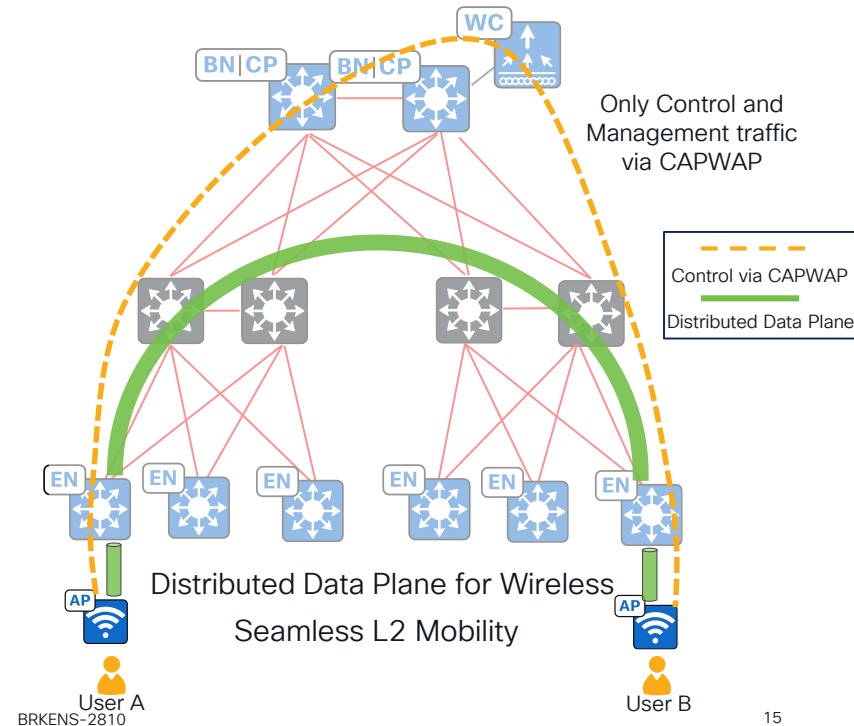
Tailormade for Campus Networks

Seamless and secure mobility of endpoints (Wireless use-case)

Traditional Network



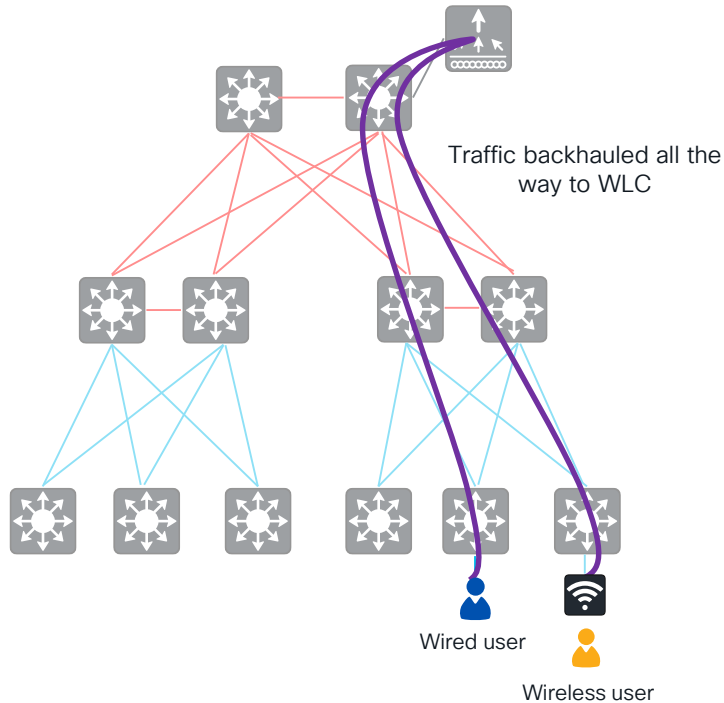
Software-Defined Access



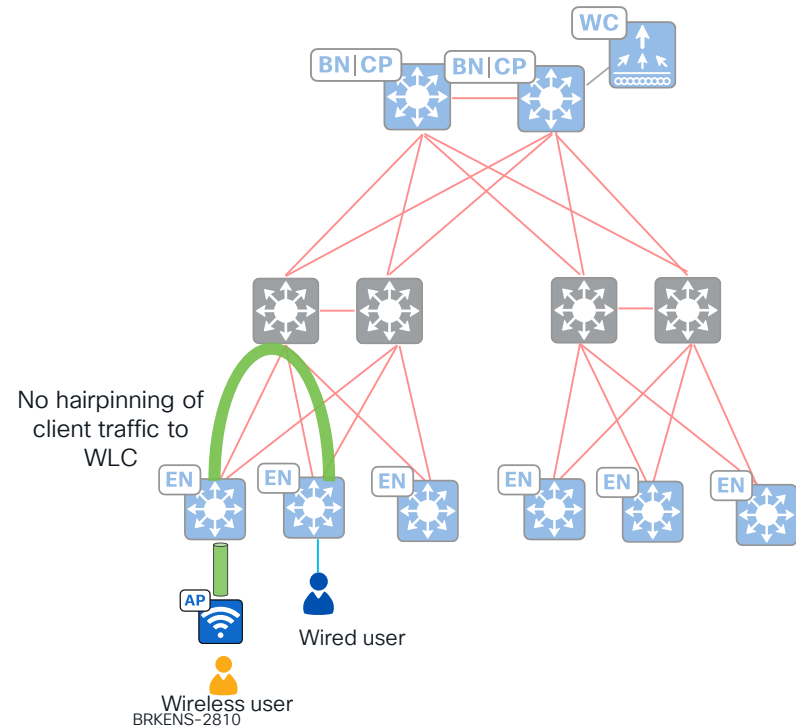
Tailormade for Campus Networks

Example of traffic flow between wireless and wired endpoint

Traditional Network



Software-Defined Access

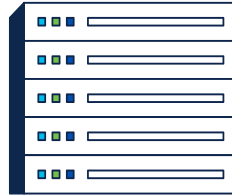


AI-Driven Insights and Telemetry

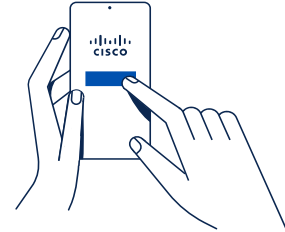
The more you know!



End user insights



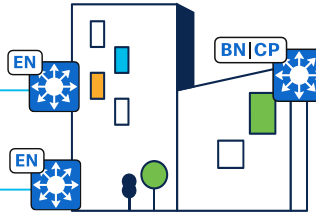
Network health and status



Application Visibility & Performance



Onboarding, Connectivity, IPv4/v6,
Device Type, MAC, VLAN, Trust Score etc.



Site Health, System Health, Topology,
Issues and Suggested Actions

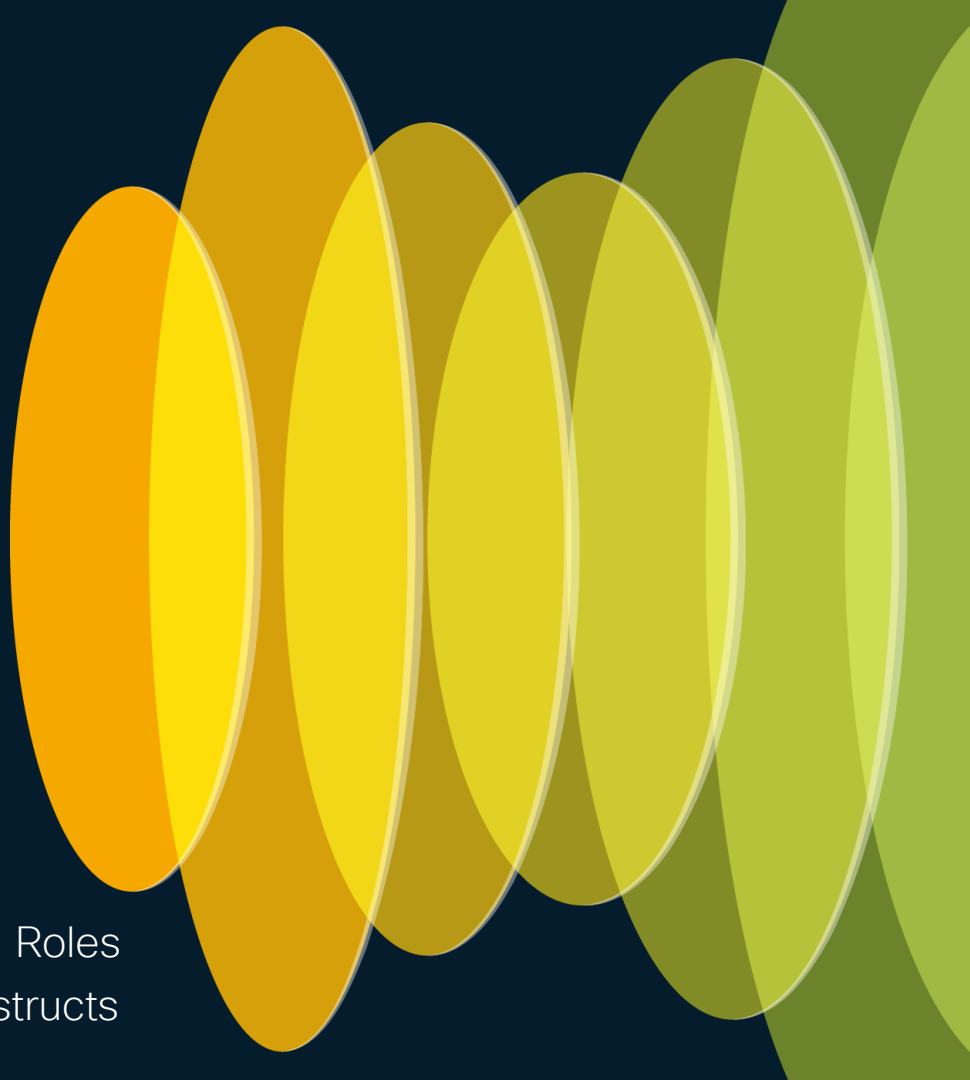


Data Center / Cloud Hosted Apps

Application Usage/Throughput, Business Relevant
Application Health, Integrations, Trends

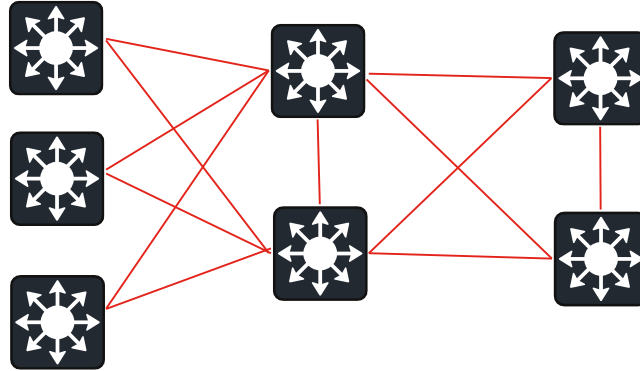
Roles and Terminology

1. Concepts
2. SD-Access Roles
3. Fabric Constructs



Underlay and Overlay

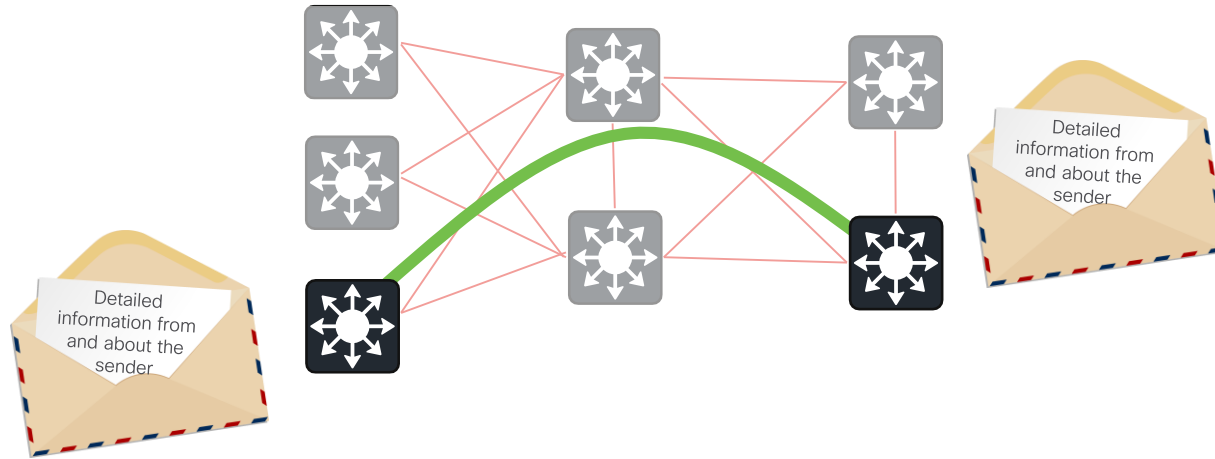
Generally Speaking



Underlay Network = Physical Infrastructure to provide IP reachability with redundancy and resiliency.

Underlay and Overlay

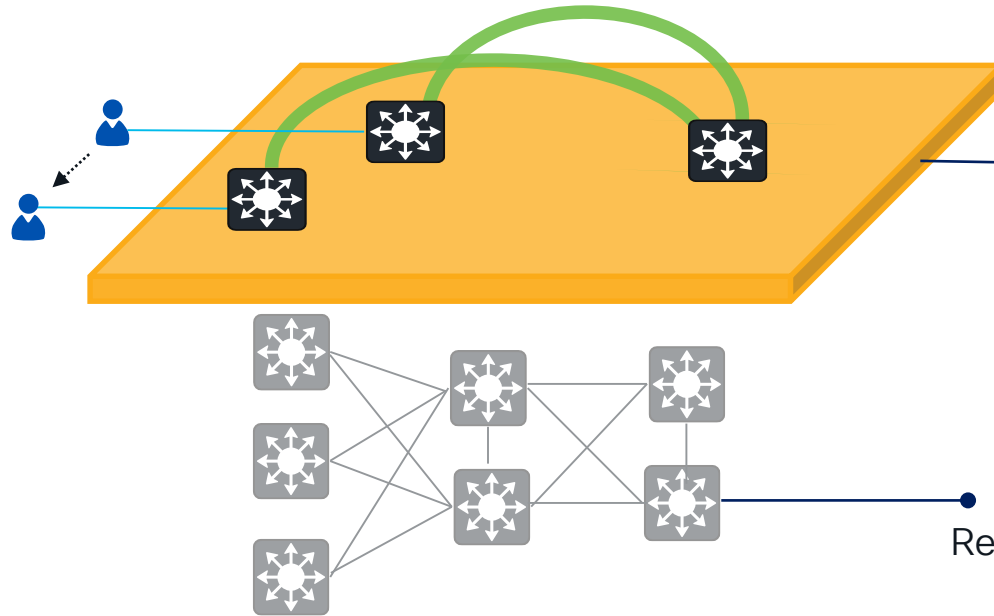
Generally Speaking



Overlay Network = Logical topology used to virtually connect devices to provide additional services, not delivered by the Underlay.

Underlay and Overlay

But why?

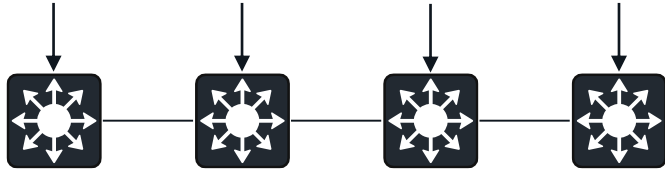


Overlay (Fabric): Flexible, Scalable and Extensible. Easy to add/modify services. Optimizes mobility events.

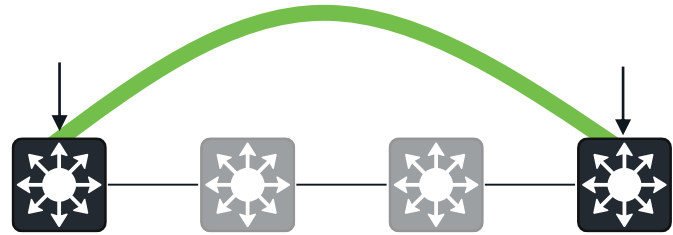
Underlay: Build and forget!
Reliable, manageable, and simple.

Underlay and Overlay

But why?



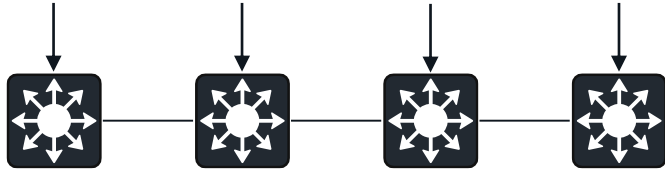
Would you rather configure
network segmentation
hop-by-hop?



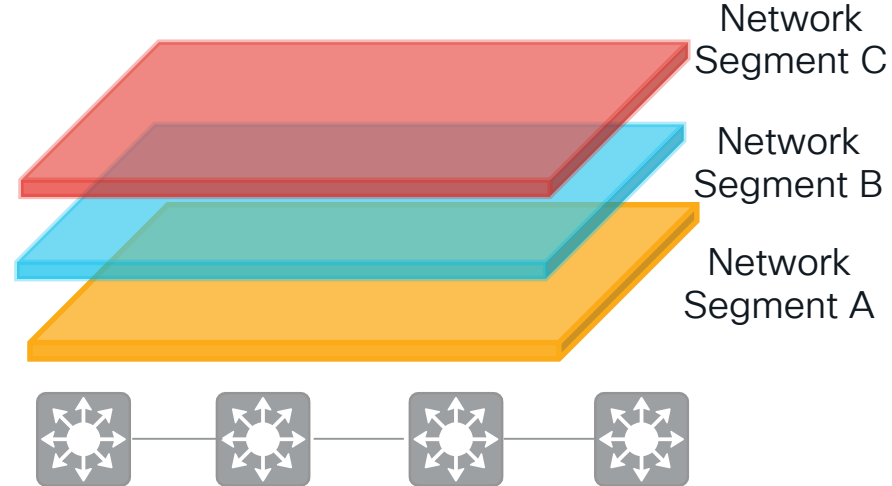
Or simply carry the
segmentation tags in the
overlay?

Underlay and Overlay

But why?



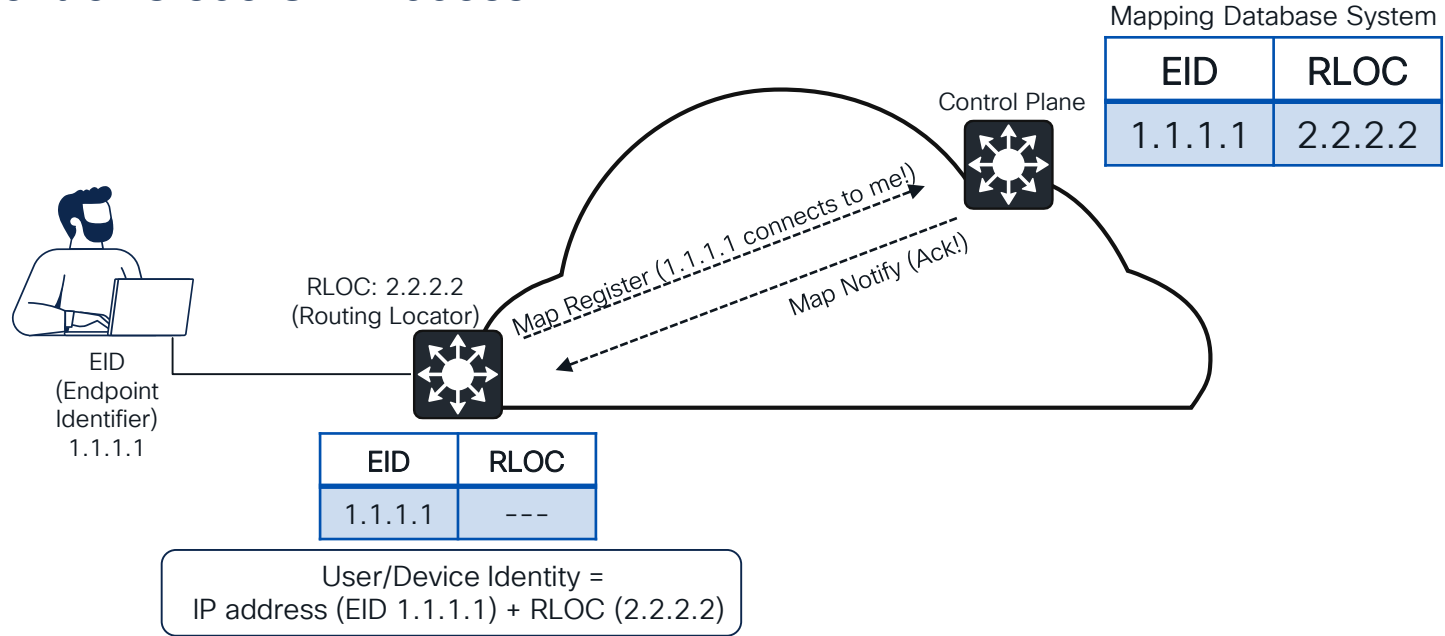
Would you rather configure
network segmentation
hop-by-hop?



Multiple segments in the overlay
that underlay is unaware of!

Underlay and Overlay

In context of Cisco SD-Access

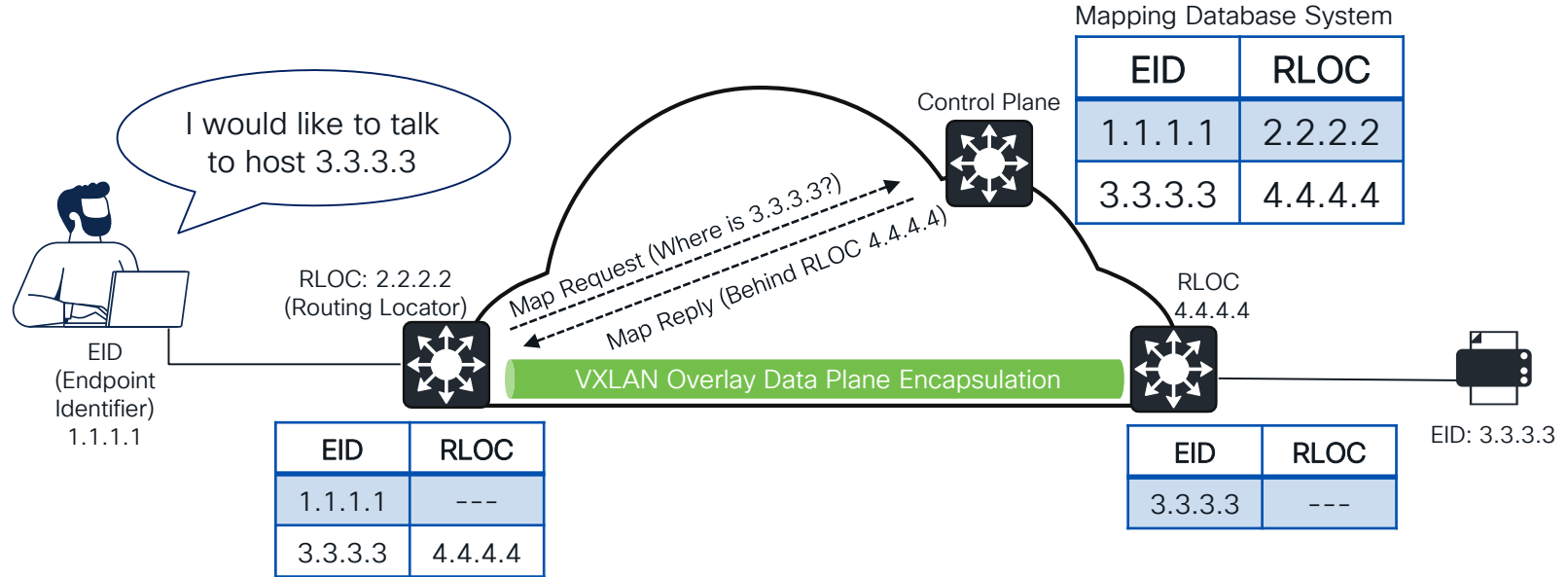


Locator Identity Separation Protocol

Control Plane Protocol of choice. Lightweight, Extensible, and Scalable.
Supports Layer 3 Overlay. Pull Based Model. Scoped Signaling.

Underlay and Overlay

In context of Cisco SD-Access



Virtual **eX**tensible **L**ocal **A**rea **N**etwork
Data Plane Protocol of choice. Supports Layer 3 and Layer 2 Overlay.

Underlay and Overlay

In context of SD-Access

What about our good friend Underlay?

A



IGP of your choice
that gets information
from Source RLOC to
Destination RLOC,
the best way it can.

B

What is an SD-Access Fabric Site?

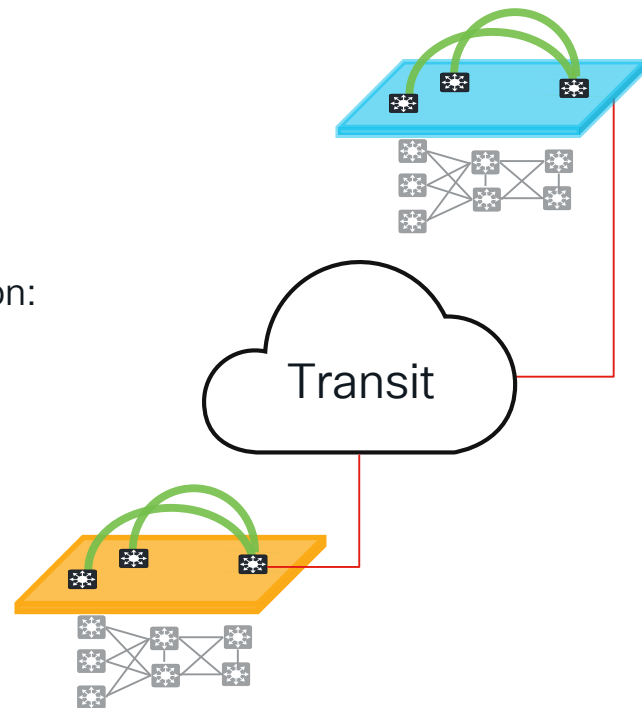
Definition and more

SD-Access Fabric Site offers programmable overlays for wired and wireless campus networks, enabled on a single physical infrastructure.

A single fabric site could be demarcated and defined based upon:

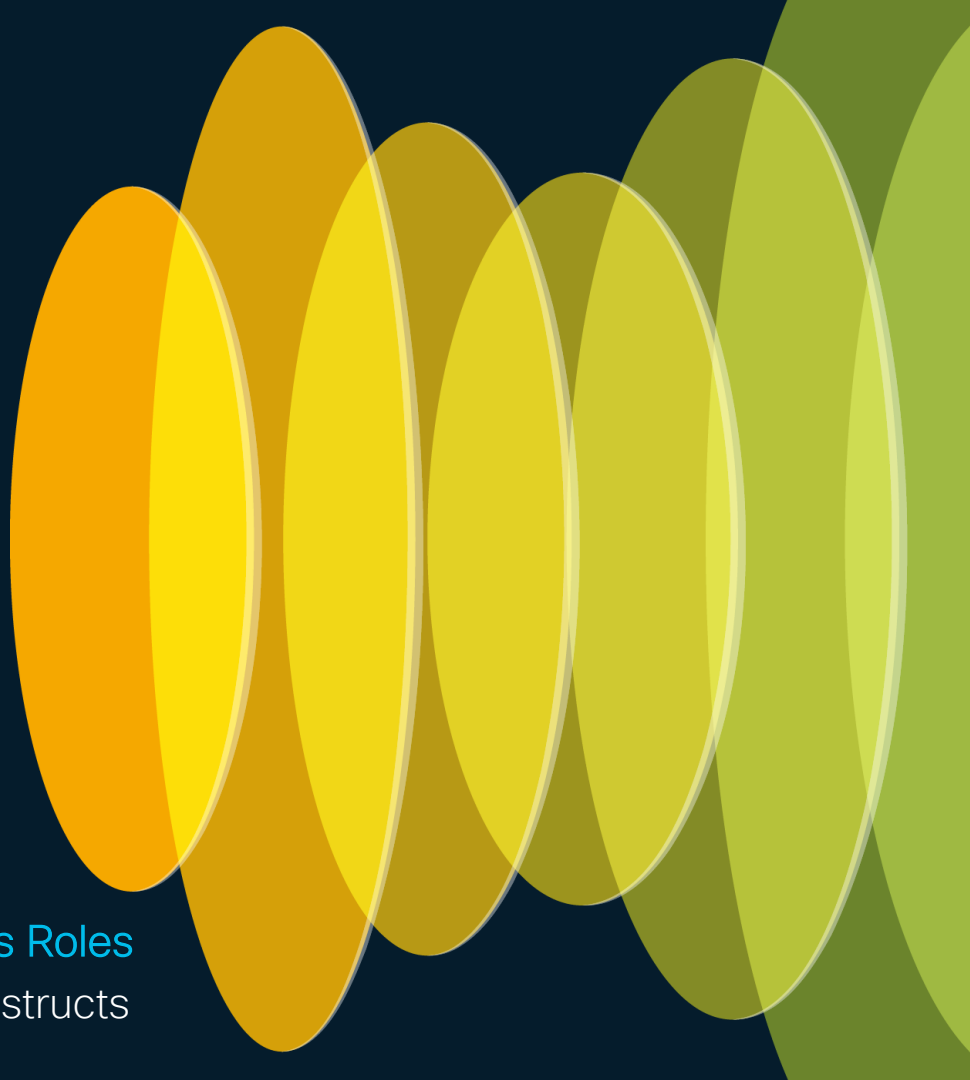
- Geographical location.
- Endpoint scale.
- Failure domain scoping.
- RTT.
- Underlay connectivity attributes.

Typically interconnected by a “Transit”.



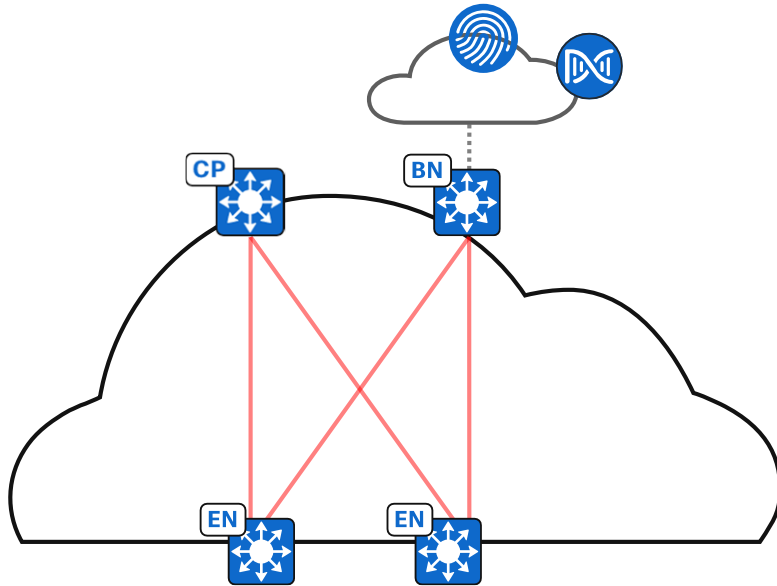
Roles and Terminology

1. Concepts
2. SD-Access Roles
3. Fabric Constructs



Cisco SD-Access Roles

Key Roles for a Complete Wired and Wireless Campus Experience



Cisco Catalyst Center

GUI and APIs for intent-based automation of wired and wireless fabric devices.



Identity Service Engine

NAC and ID services for dynamic endpoint to Security Group Tag mapping and policy distribution.



Control Plane Node

Map System that tracks endpoint to fabric node relationships.



Border Nodes

Connects external L3 and L2 networks to the Cisco SD-Access fabric.

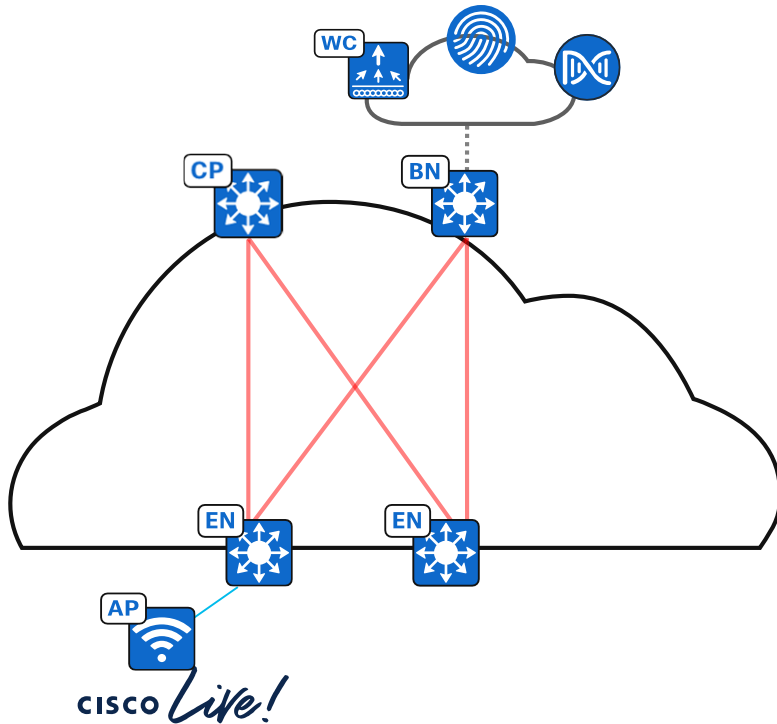


Edge Nodes

Connects wired endpoints to the Cisco SD-Access fabric and optionally enforces micro-segmentation policy.

Cisco SD-Access Roles

Key Roles for a Complete Wired and Wireless Campus Experience



Control Plane Node

Map System that tracks endpoint to fabric node relationships.



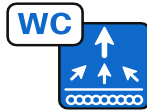
Border Nodes

Connects external L3 and L2 networks to the Cisco SD-Access fabric.



Edge Nodes

Connects wired endpoints and Fabric APs to the Cisco SD-Access fabric and optionally enforces micro-segmentation policy.



Fabric Wireless Controller

Fabric WLC is integrated into the SD-Access Control Plane (LISP) communication.



Fabric Access Point

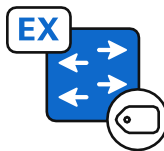
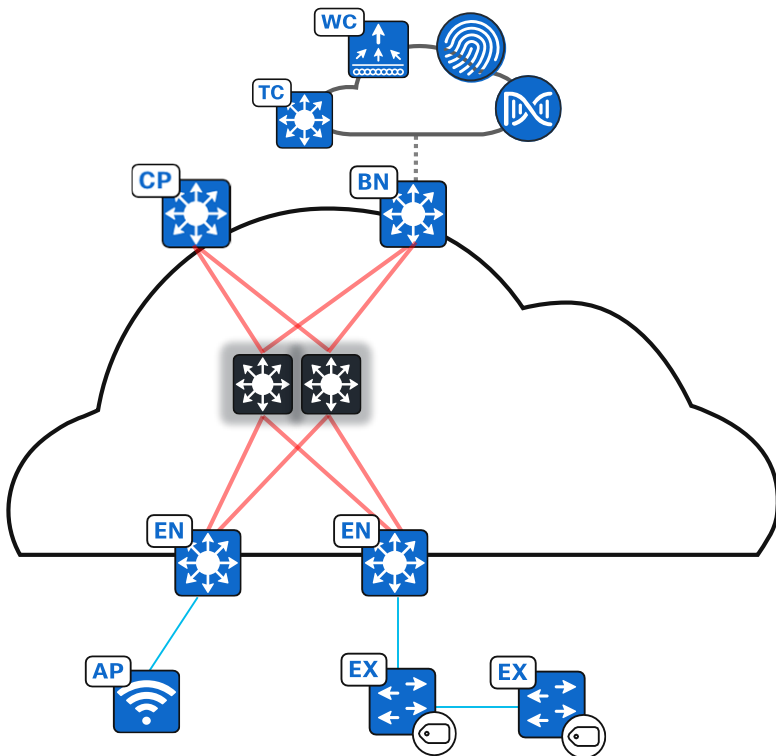
Switches endpoint traffic to the adjacent Edge Node.



Additional
Information

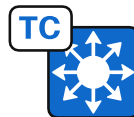
Cisco SD-Access Roles

Additional Roles for Reference



Extended Nodes

A switch operating at Layer 2 that extends fabric connectivity and optionally enforces micro-segmentation policy.



Transit Control Plane Nodes

Facilitates connectivity of multiple SD-Access fabric sites while preserving end to end segmentation.



Intermediate Nodes

Moves data between fabric nodes. Can be one or many hops. Part of the underlay.



Additional
Information

Cisco SD-Access Roles

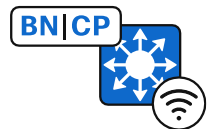
Some of the Supported Colocations



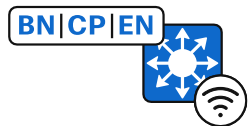
Border Node and Control Plane Node.



Border Node, Control Plane Node, and Fabric Edge Node.



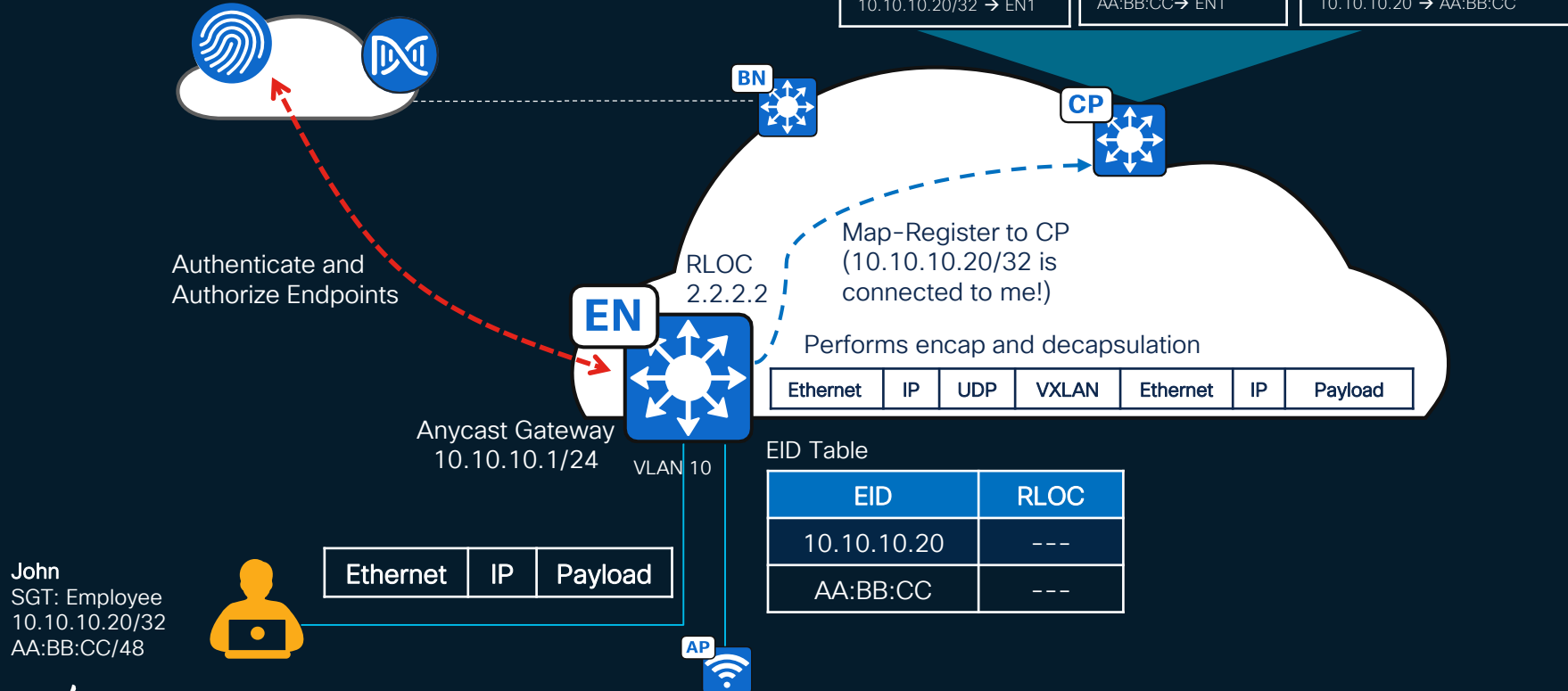
Border Node, Control Plane Node, and Embedded Wireless Controller.



Border Node, Control Plane Node, Fabric Edge Node, and Embedded Wireless Controller.

Edge Node

Provides First Hop Services for Endpoints



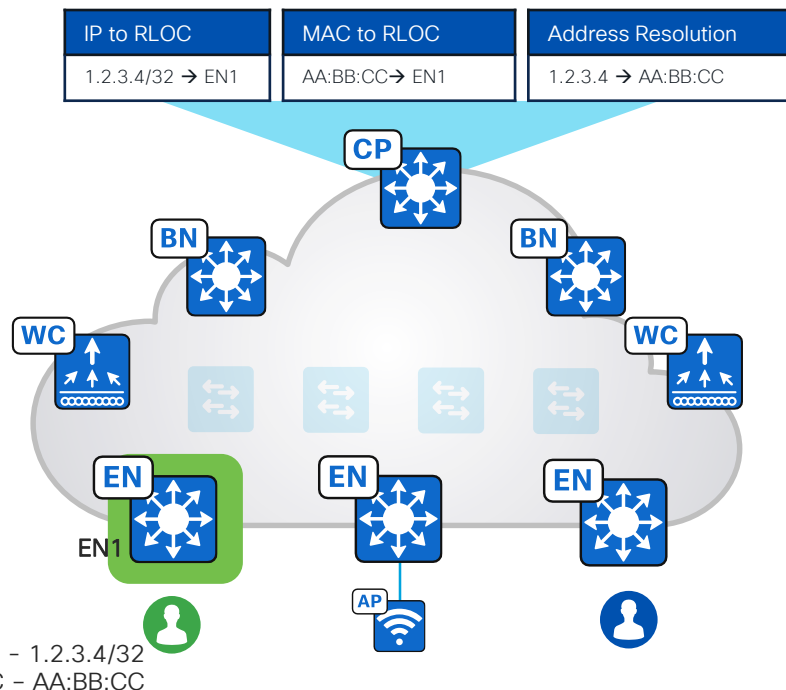


Additional
Information

Cisco SD-Access Fabric

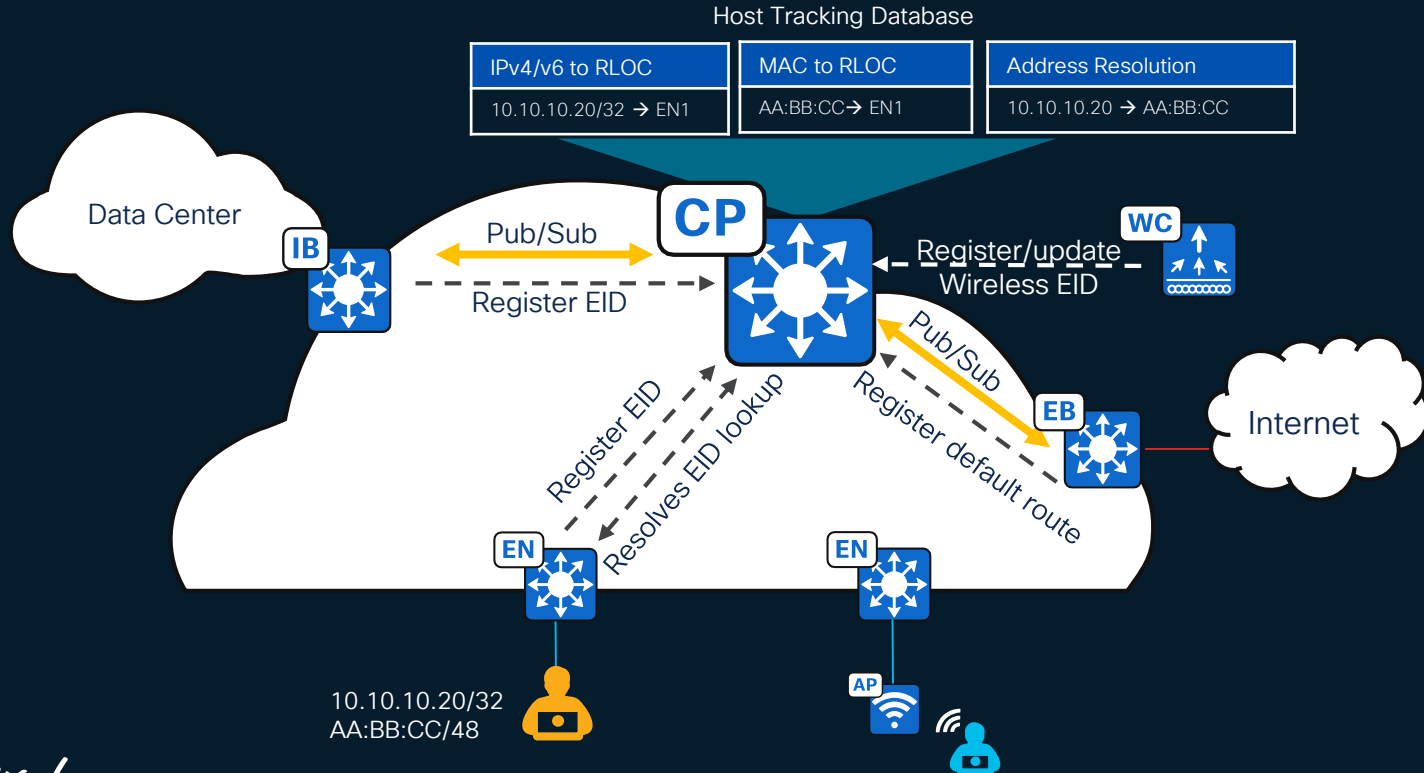
Edge Node Provides First Hop Services for Endpoints

- Responsible for Authenticating and Authorizing endpoints (e.g. 802.1X, MAB, static) in concert with ISE.
- Register Endpoint IDs (IPv4, IPv6, MAC) with the Control Plane Nodes.
- Provide an Anycast Gateway for the connected wired and wireless endpoints.
- Performs VXLAN encapsulation and decapsulation of traffic to and from all connected wired endpoints.



Control Plane Node

Maintains Host Tracking Database containing Endpoint Attribute to Location Mapping



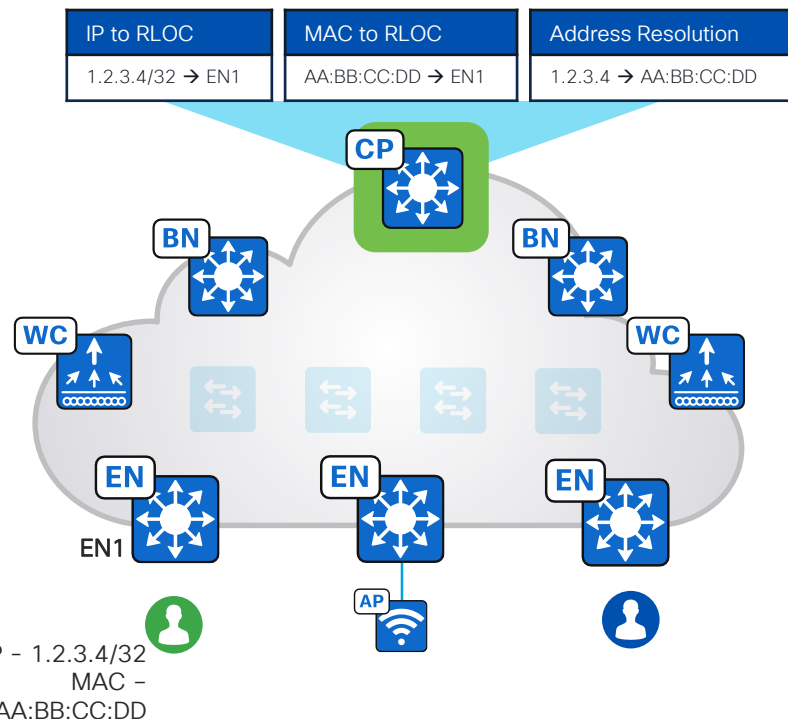


Additional
Information

Cisco SD-Access Fabric

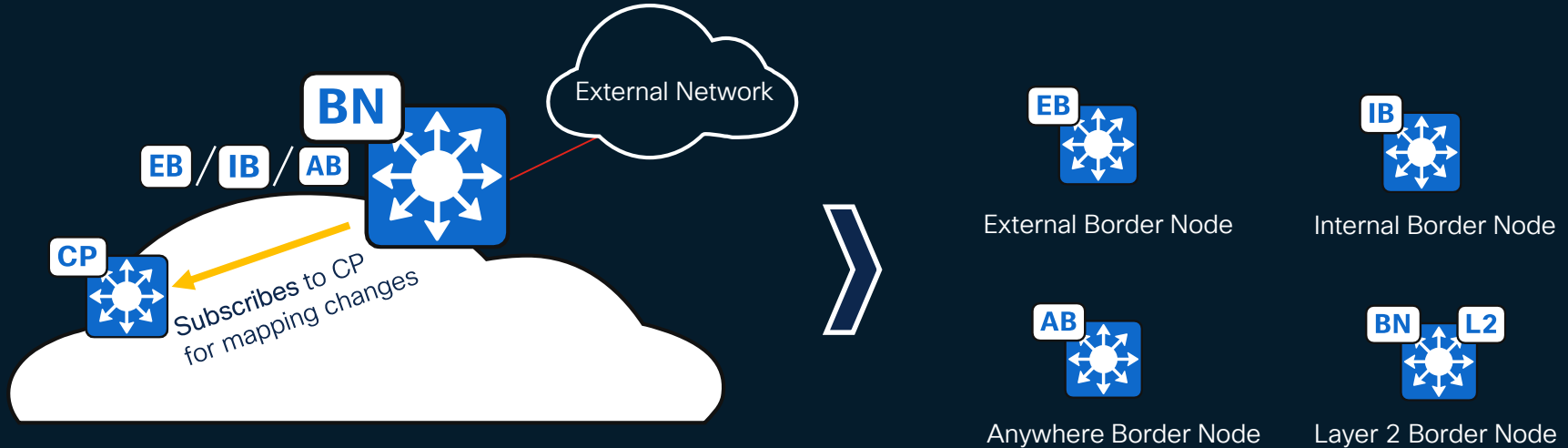
Control Plane Node Maintains a Host Tracking Database to Map Location Information

- A simple Host Database that maps Endpoint IDs to locations, along with other attributes.
- Host Database supports multiple types of Endpoint ID lookup types (IPv4, IPv6 or MAC).
- Receives Endpoint ID map registrations from Edge Nodes, Border Nodes and Fabric Wireless LAN Controllers.
- Resolves lookup requests from Edge Nodes and Border Nodes, to locate destination Endpoint IDs.
- Publishes registrations to Subscribers (Border Nodes).



Border Node

Gateway between the SD-Access fabric site and the networks external to the fabric

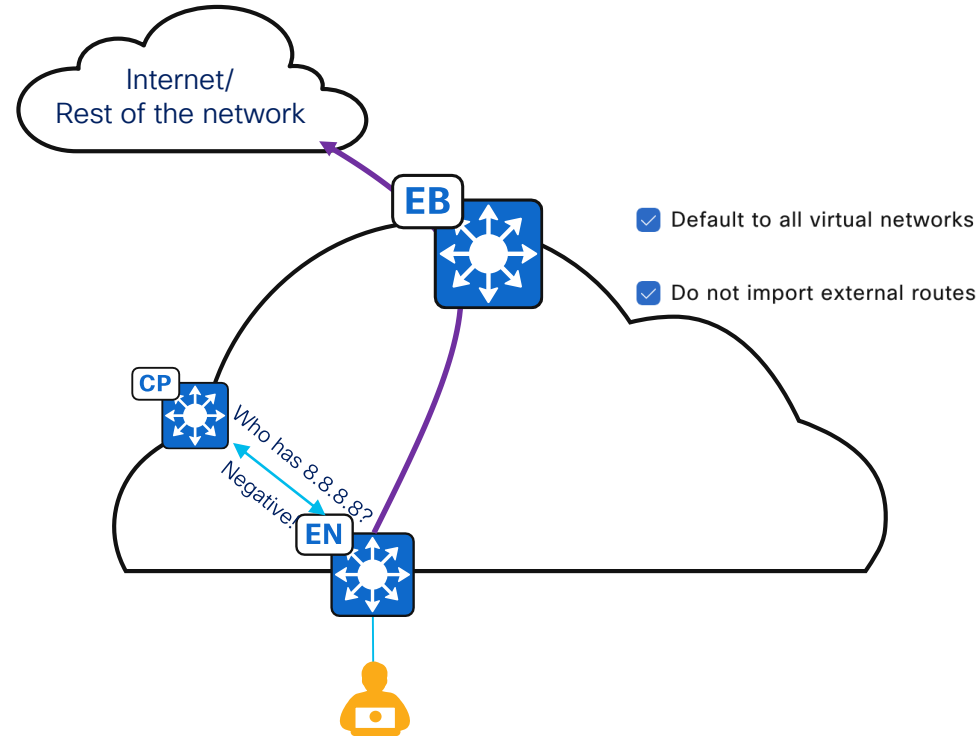


Border Nodes connect external L3 and L2 networks to the Cisco SD-Access fabric.
As a result, performs VXLAN encapsulation and decapsulation.
There are 4 types of Border Nodes.

Cisco SD-Access Fabric

External Border Node

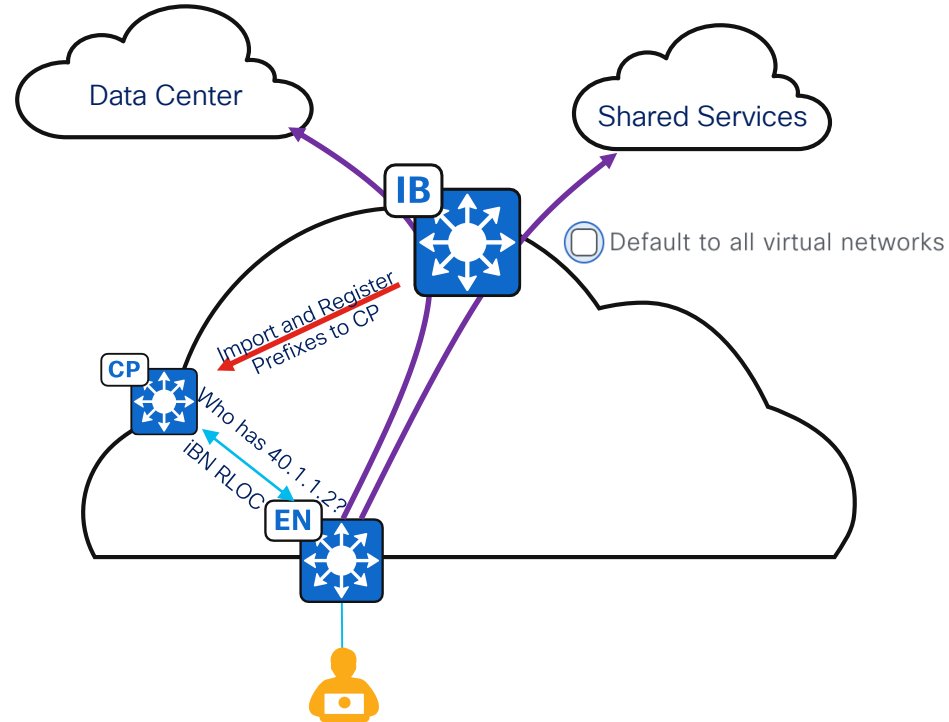
- The most common configuration.
- Exports all fabric subnets to outside the Fabric Site as eBGP summary routes.
- Does not register IP prefixes from outside the Fabric Site into the fabric Control Plane.
- Acts as a gateway of last resort for the Fabric Site.



Cisco SD-Access Fabric

Internal Border Node

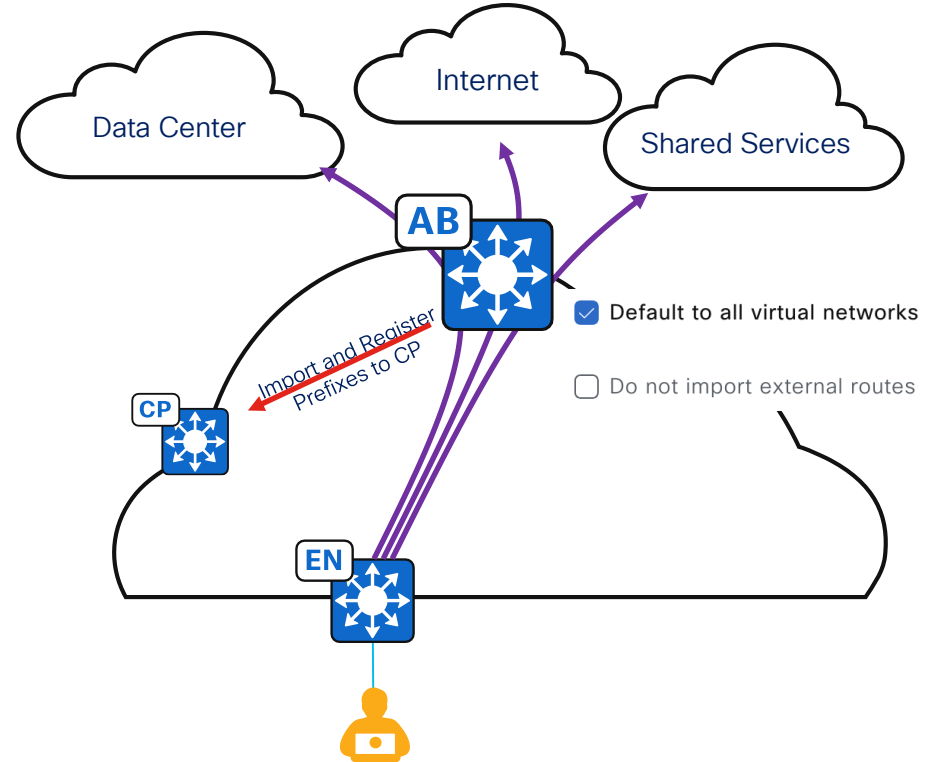
- Exports all fabric subnets to outside the Fabric Site as eBGP summary routes.
- Imports and registers eBGP-learned IPv4/IPv6 prefixes from outside the Fabric Site, into the fabric Control Plane.
- Does not act as a gateway of last resort for the Fabric Site.



Cisco SD-Access Fabric

Internal + External Border Node

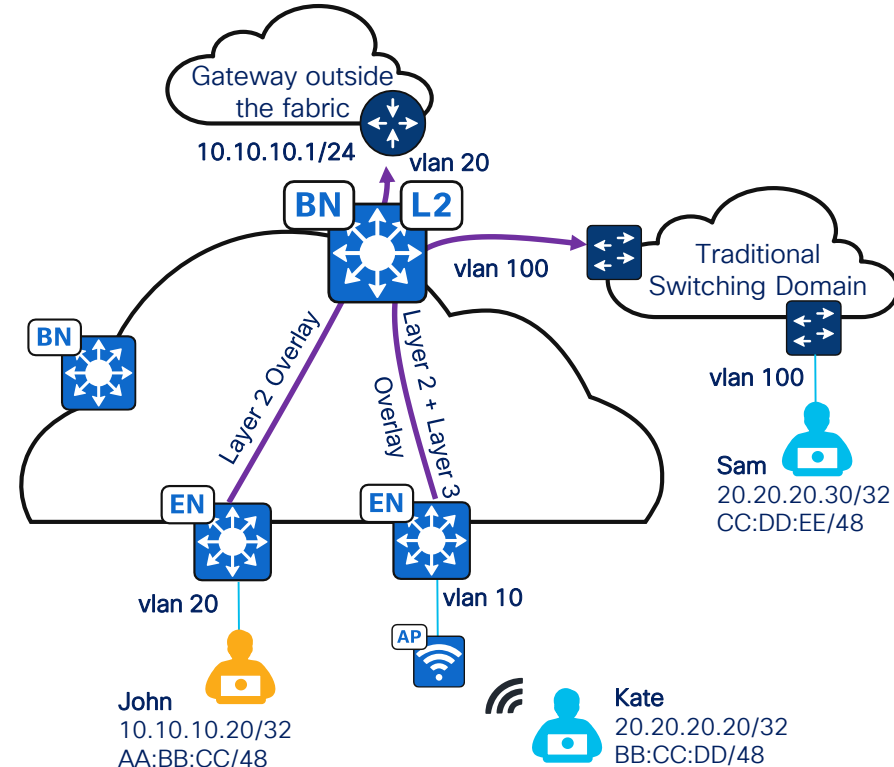
- Exports all fabric subnets to outside the Fabric Site as eBGP summary routes.
- Imports and registers eBGP-learned IPv4/IPv6 prefixes from outside the Fabric Site, into the fabric Control Plane.
- Acts as a gateway of last resort for the Fabric Site.



Cisco SD-Access Fabric

Layer 2 Border Node

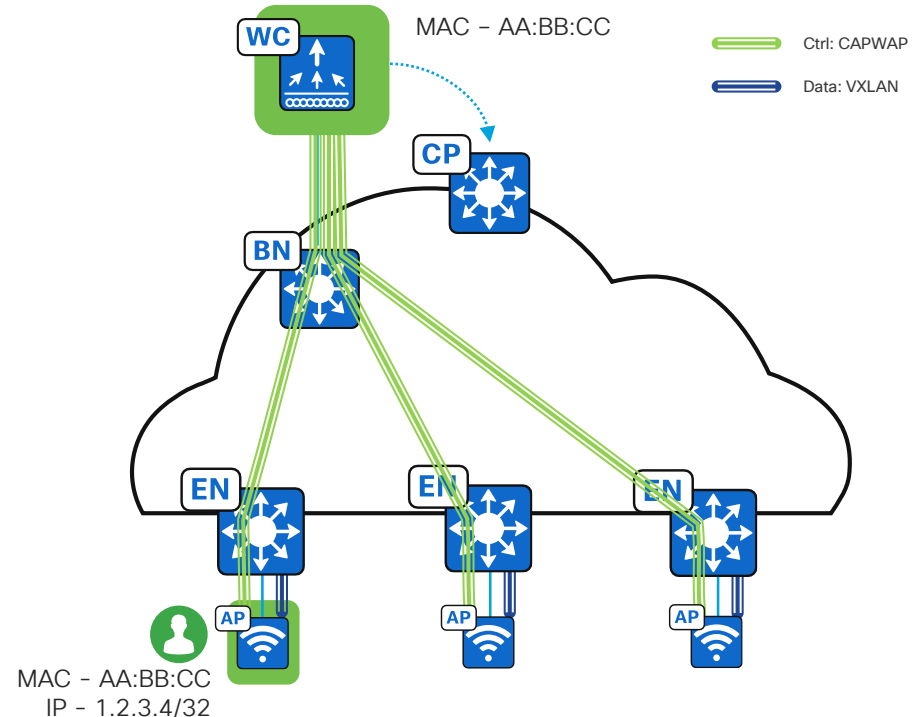
- Acts as Layer 2 handoff for pure Layer 2 Overlays or Layer 2 + Layer 3 Overlays.
- Allows VLAN translation between SD-Access network segments and non-fabric VLAN IDs.
- Dual homing requires link aggregation; STP is not tunneled within the SD-Access Fabric.
- Ideally should be separate device from the Layer 3 Border Node.



Cisco SD-Access Fabric

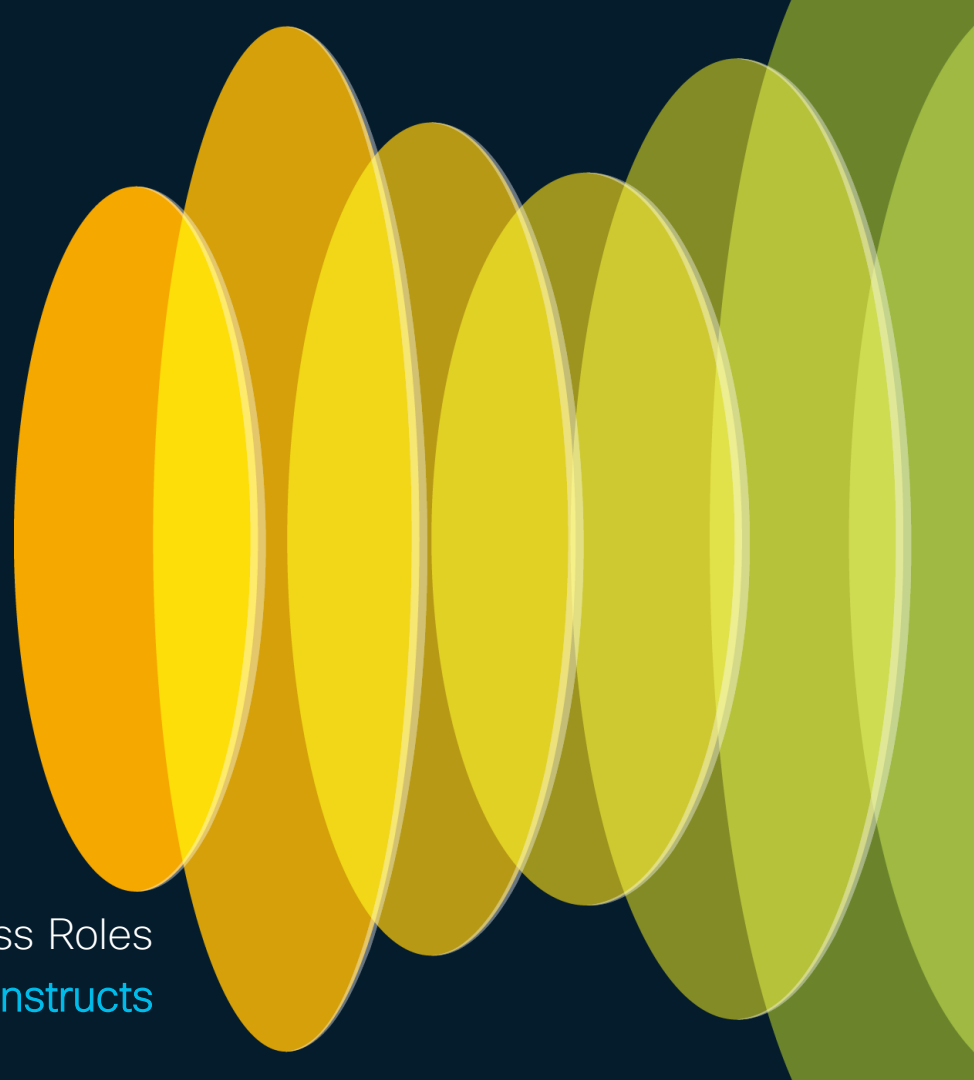
Fabric Enabled Wireless Unifies Wired and Wireless Management, Policy and Data Planes

- Fabric WLC accessible through a Fabric Border Node (Underlay). Can be several hops away.
- Fabric Enabled APs reside in a dedicated IP range and communicate with the WLC (CAPWAP Control).
- Fabric WLC registers endpoints with the Control Plane Node.
- Fabric APs switch endpoint traffic to the adjacent Edge Node.
- Wireless endpoints use same data plane and policy plane as wired endpoints.



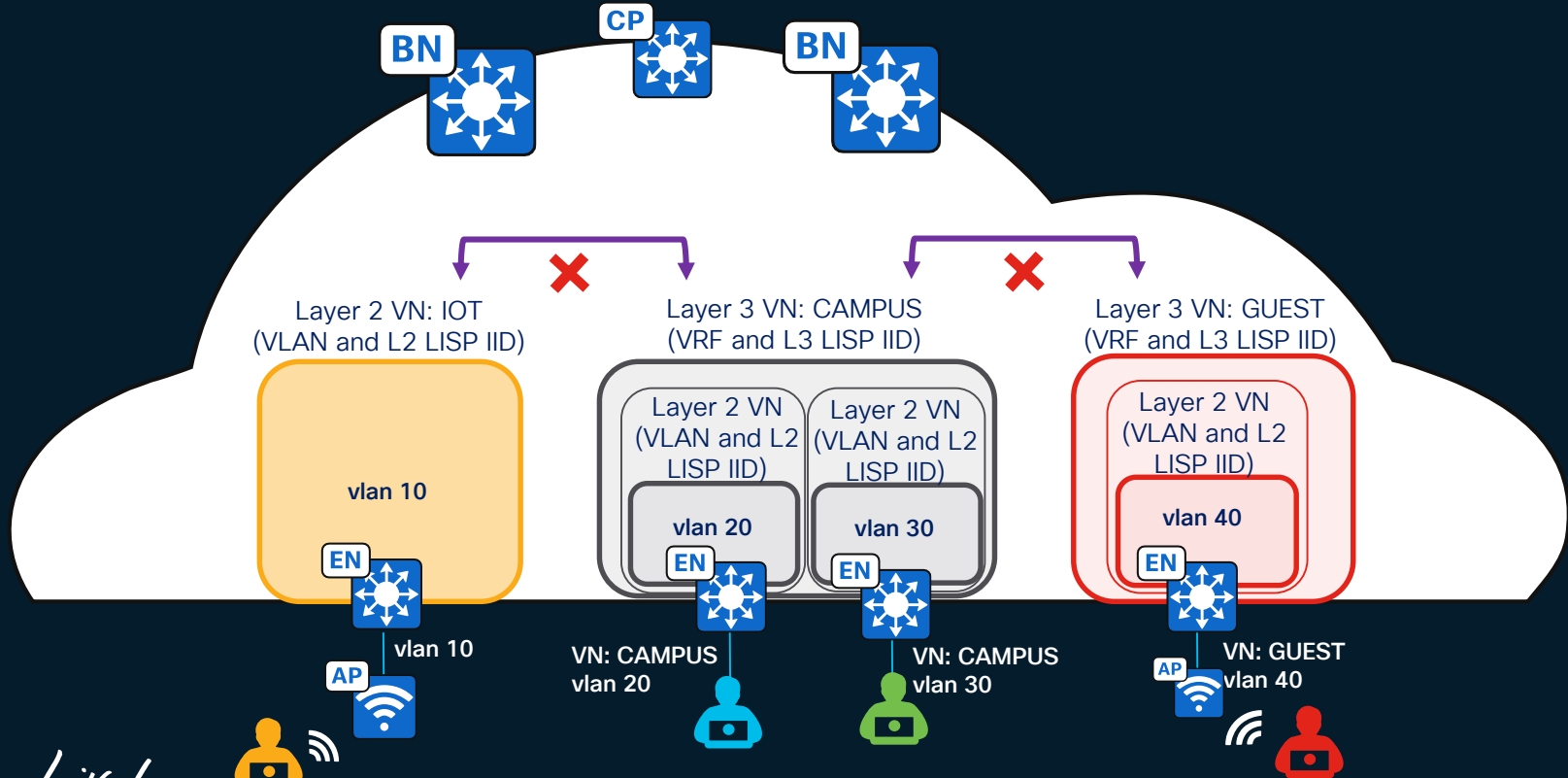
Roles and Terminology

1. Concepts
2. SD-Access Roles
3. Fabric Constructs



Virtual Networks

Overlay Network Virtualization, also known as Macro-segmentation.



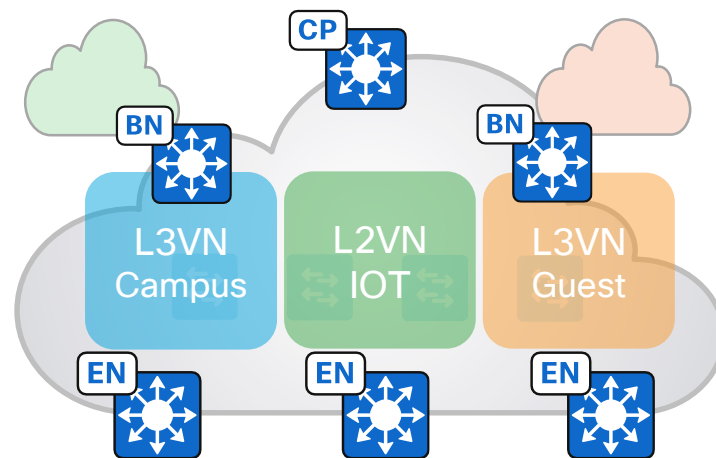


Additional
Information

Cisco SD-Access Fabric

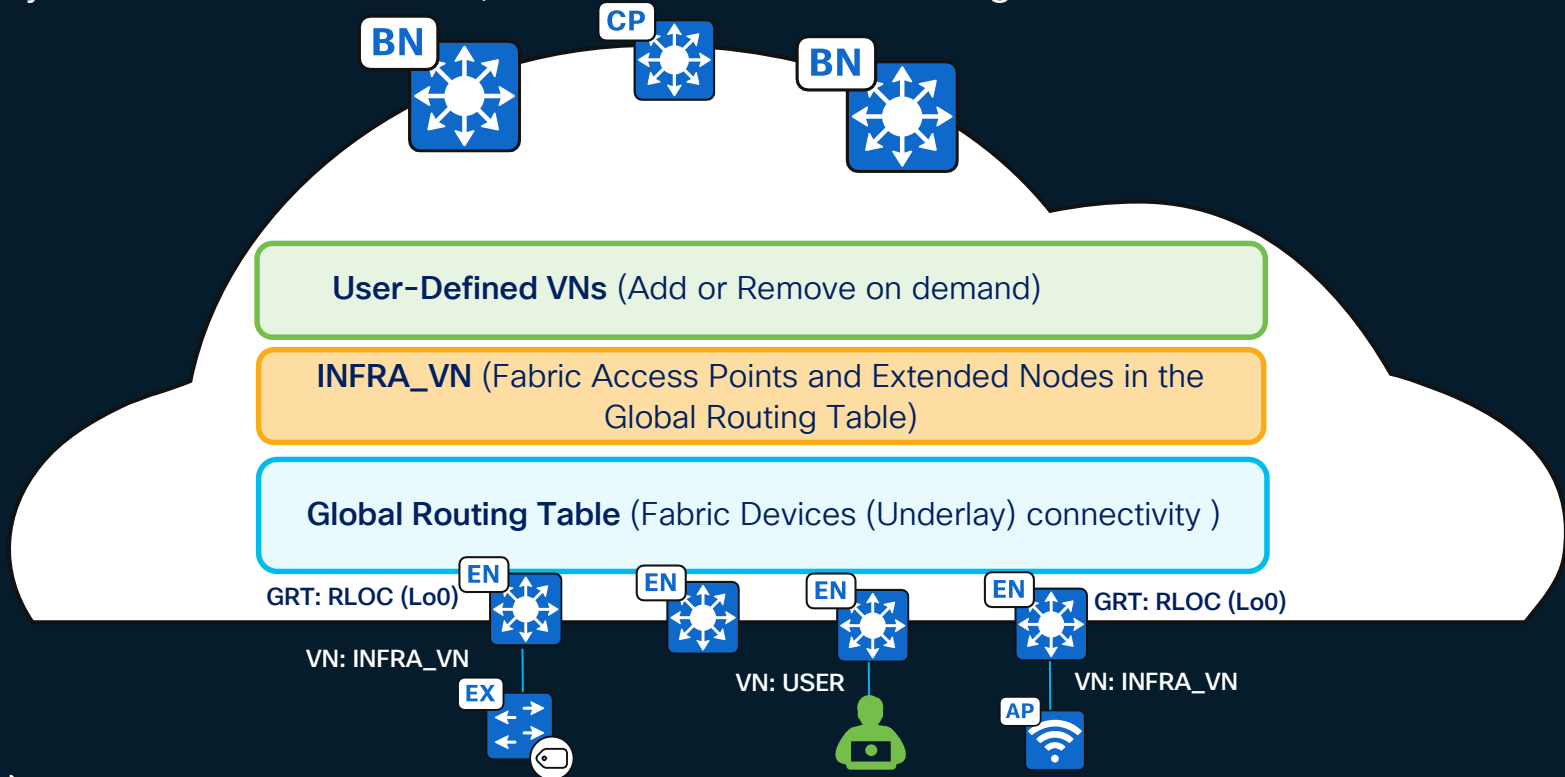
Virtual Networks

- Layer 3 Virtual Networks use VRFs and LISP Instance IDs to maintain separate routing topologies.
 - Endpoint IDs (IPv4/IPv6 addresses) are routed within an L3VN.
- Layer 2 Virtual Networks use LISP Instance IDs and VLANs to maintain separate switching topologies.
 - Endpoint IDs (MAC addresses) are switched within an L2VN.
- Edge Nodes, Border Nodes and Fabric APs add a VNID (the LISP IID) to the fabric encapsulation.



Layer 3 Virtual Networks

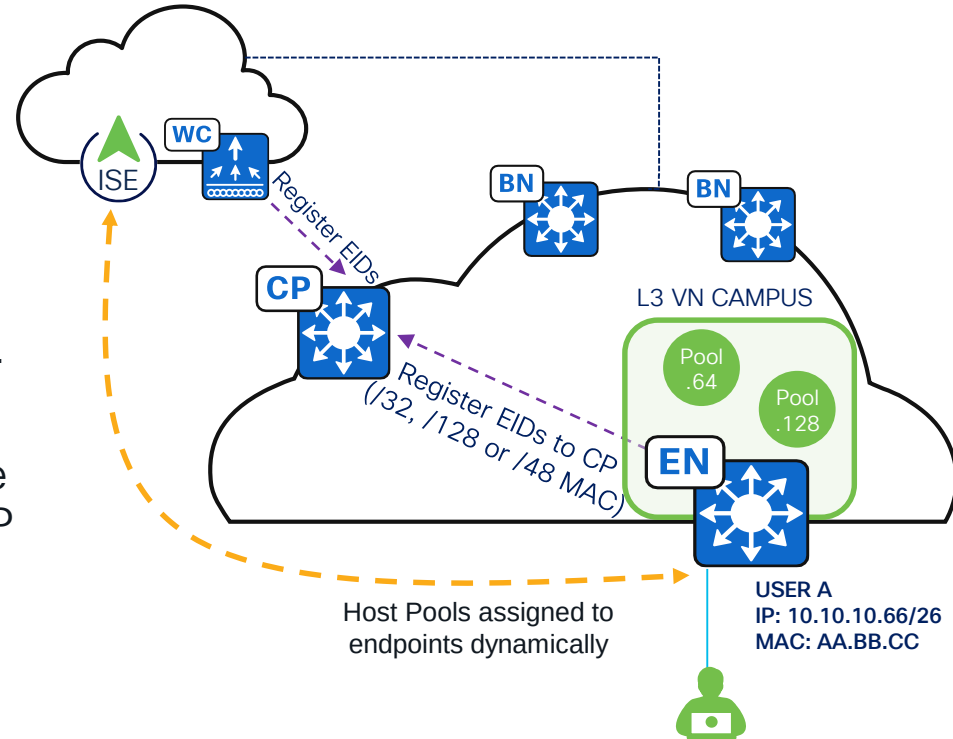
Overlay Network Virtualization, also known as Macro-segmentation.



Cisco SD-Access Fabric

Host Pools Provide a Default Gateway and Basic IP Services for Endpoints

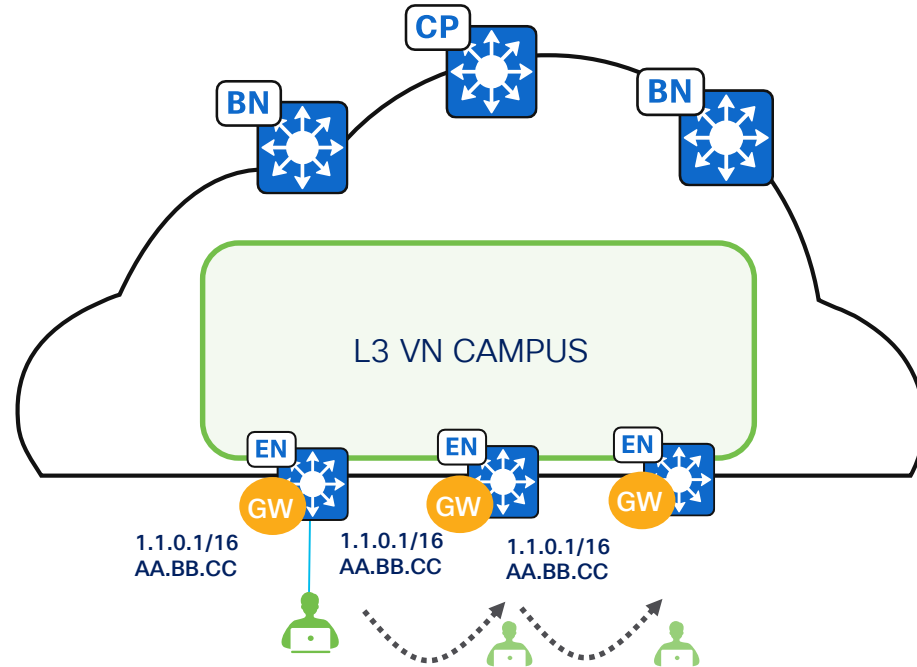
- Edge Nodes instantiate an access VLAN and a Switched Virtual Interface (SVI) with user-defined IPv4/IPv6 addresses per Host Pool.
- Host Pools assigned to endpoints dynamically by AAA or statically per port.
- Edge Nodes and Fabric WLCs register endpoint IDs (/32, /128 or MAC) with the Control Plane, enabling IP mobility; any IP address anywhere.



Cisco SD-Access Fabric

Anycast Gateway Provides a Default Gateway for IP-Capable Endpoints

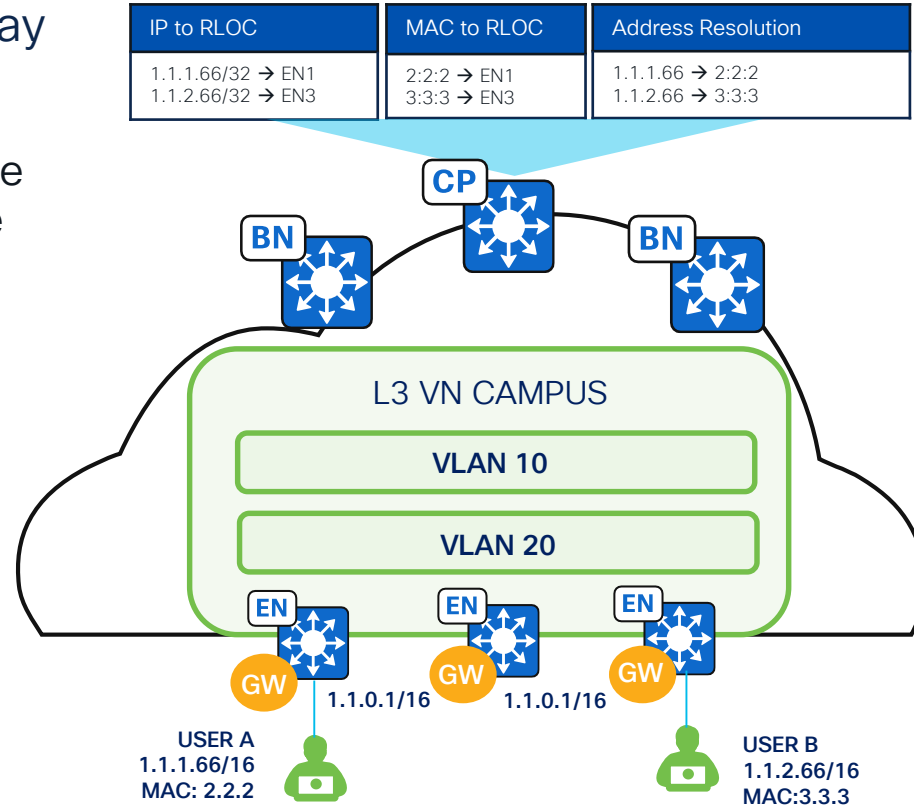
- Similar principle and behavior to FHRP with a shared virtual IPv4/IPv6 addresses and MAC address.
- The same Switch Virtual Interface (SVI) is present on all Edge Nodes with the same virtual IP and MAC.
- The wired or wireless endpoint can connect to any switch or AP in the fabric and communicate with the same Anycast Gateway.



Cisco SD-Access Fabric

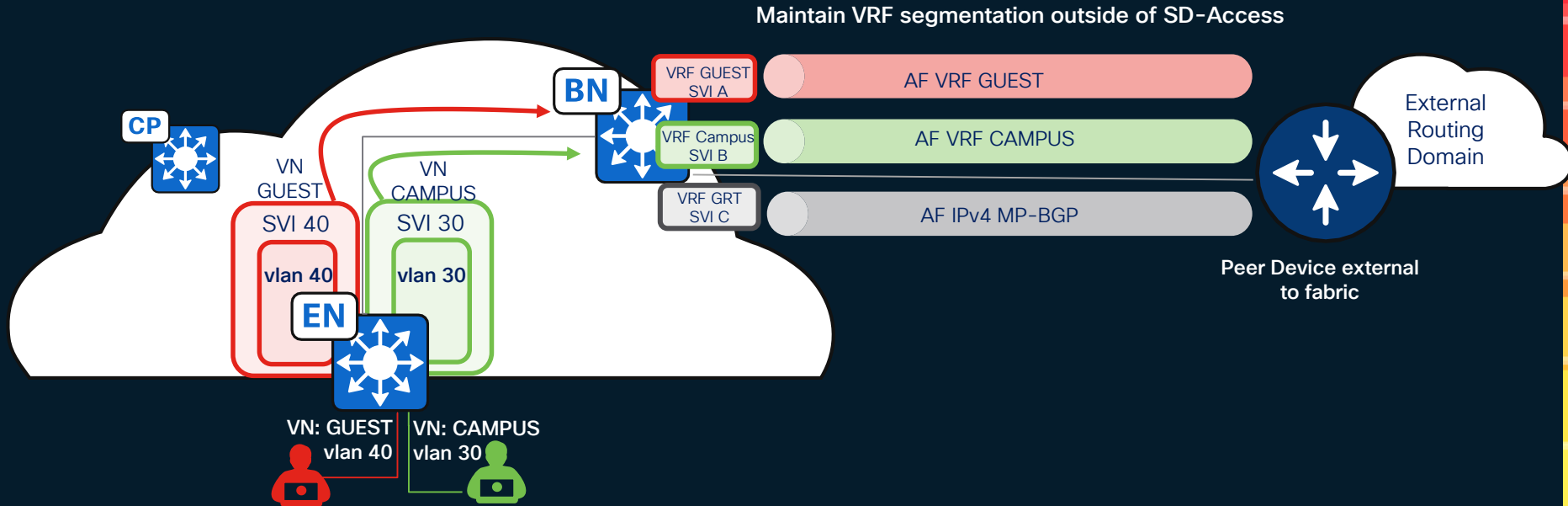
Host Pools are “stretched” via the Overlay

- Endpoint IPv4/IPv6 traffic arrives on an Edge Node and is then routed or switched by the Edge Node.
- Fabric Dynamic EID mapping allows endpoint-specific (/32, /128, MAC) advertisement and mobility.
- No longer need VLANs to interconnect endpoints across Edge Nodes, this happens in the Overlay without broadcast flooding.



Layer 3 Handoff

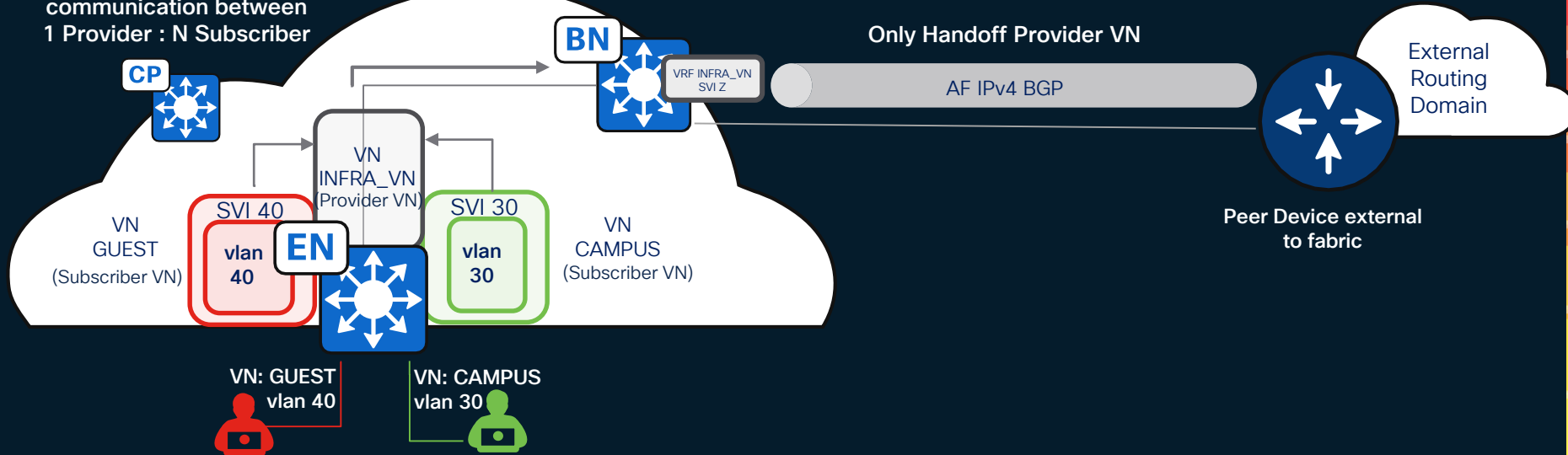
Per-Layer-3-Virtual-Network Layer 3 Handoff using Peer Device



Extranet Layer 3 Handoff

Helps achieve route-leaking natively in LISP SD-Access Fabric

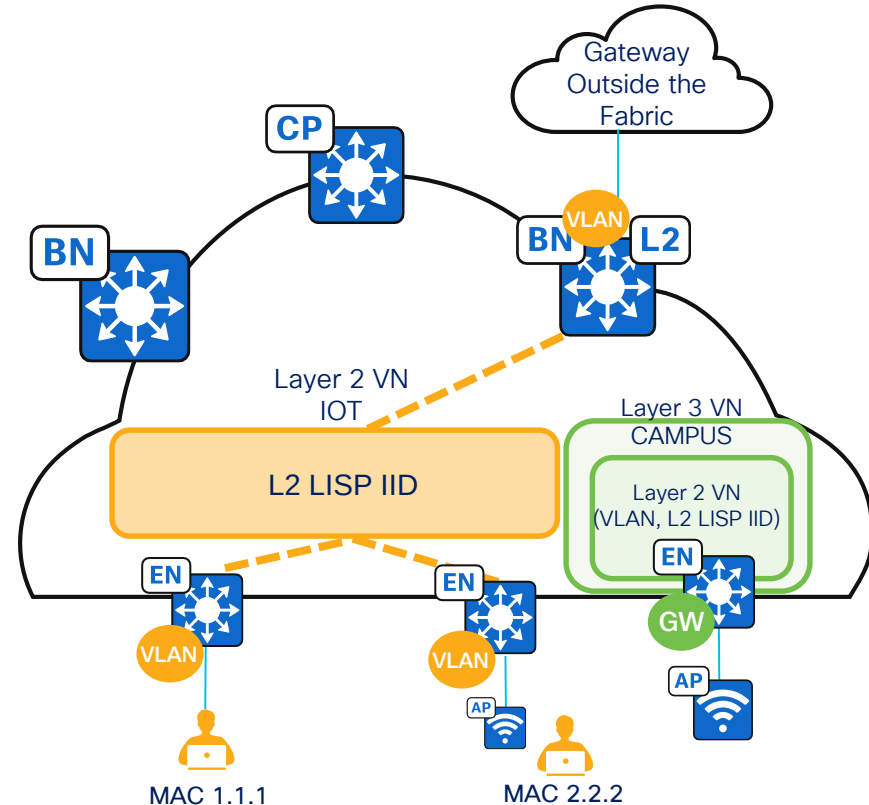
Extranet Policy to allow communication between 1 Provider : N Subscriber



Cisco SD-Access Fabric

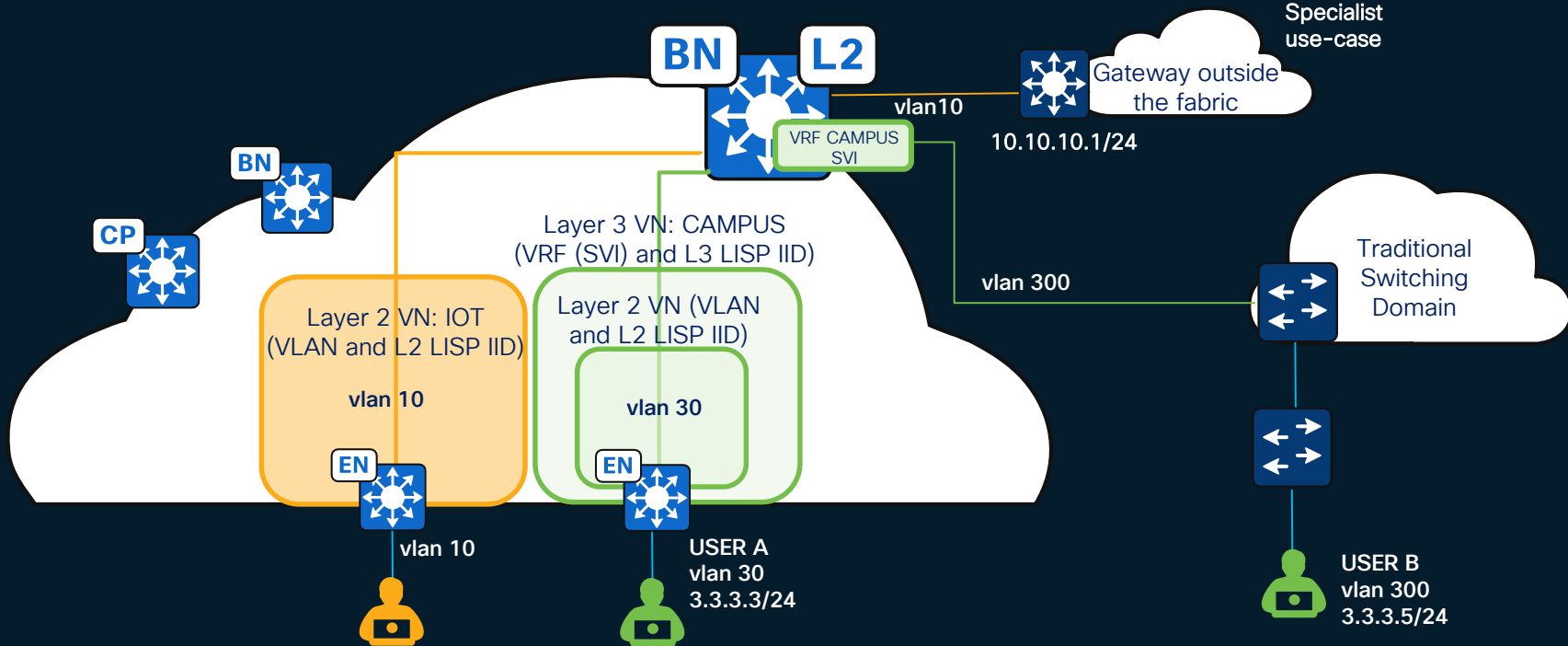
Layer 2 Virtual Networks

- By default, an L2VN is deployed with each Anycast Gateway and Layer 2 Flooding is disabled. Layer 2 Flooding can be enabled, if necessary, to service niche applications.
- L2VN can be deployed without an Anycast Gateway, and Layer 2 Flooding cannot be disabled.
- Sometimes referred to as “Gateway Outside the Fabric”.
- If Layer 2 Flooding is enabled, a Multicast Underlay P2MP tunnel is established between all Fabric Nodes.



Layer 2 Handoff

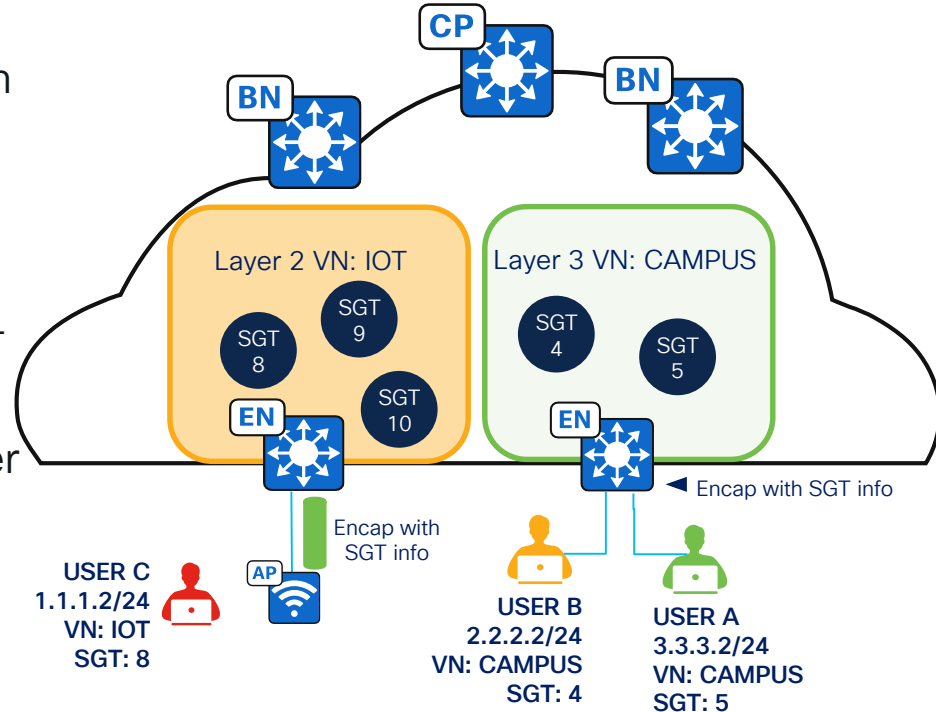
Layer 2 Virtual Networks handoff through a user-defined VLAN



Cisco SD-Access Fabric

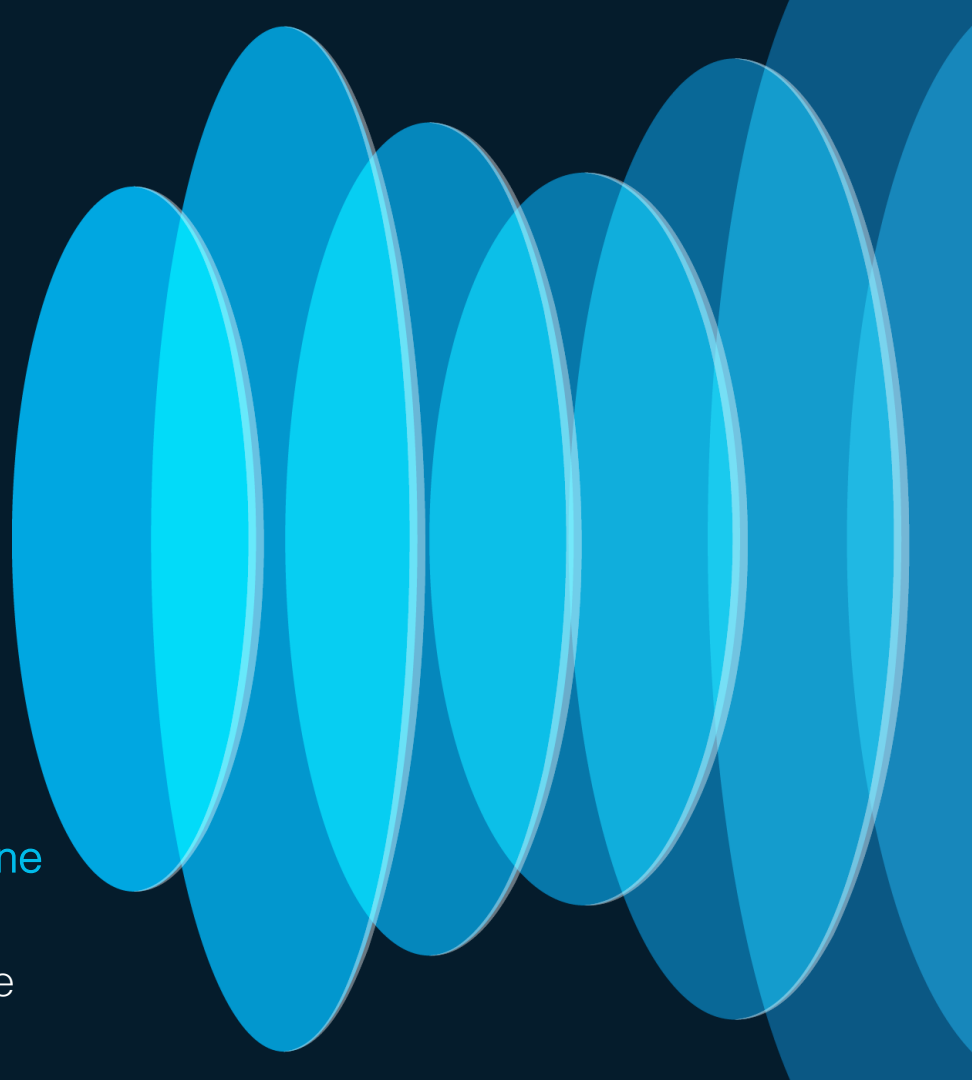
A Security Group Tag Assigns a “Group” to Each Endpoint

- Edge Nodes and Fabric APs assign a unique Scalable Group Tag (SGT) to each endpoint in concert with ISE.
- Edge Nodes and Fabric APs add an SGT to the fabric encapsulation.
- SGTs are used to implement IP-address-independent traffic policies.
- SGTs can be extended to numerous other networking technologies e.g., Cisco Secure Firewall, Cisco SD-WAN, some third-party devices, etc.



Fabric Fundamentals

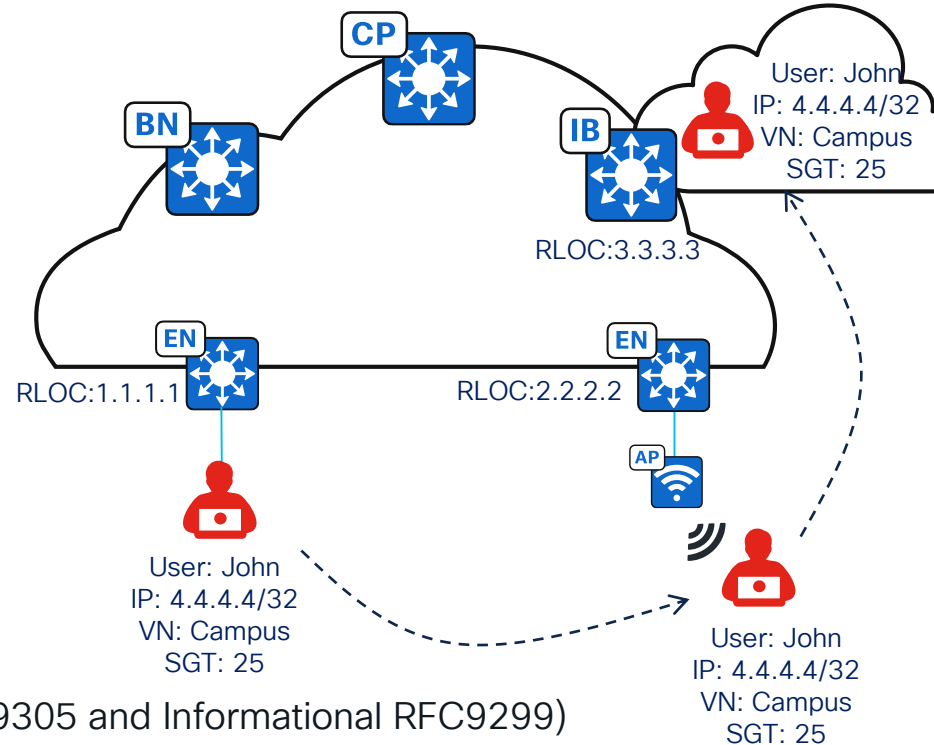
1. Control Plane
2. Data Plane
3. Policy Plane



Cisco SD-Access Fabric

Control Plane: Locator/ID Separation Protocol (LISP)

Where you are in a network can change, but **who** you are in the network remains the same.



(IETF Standards Track RFC9300-RFC9305 and Informational RFC9299)

Cisco SD-Access Fabric

Why LISP?

Scalable

Pull model • No massive routing tables • Creates 'DNS' for routing •

Conversational learning Massive Scalability • BGP and IGPs cannot scale like LISP • Purpose-built for scale (DFZ was the impetus for LISP)

Efficient

Address-Family Agnostic • Supports IPv4, IPv6, and MAC Address Families • Wired and Wireless Unification • WLC participates in LISP control plane communication. • Wired and Wireless endpoints have policy applied at same point in the network. Host Mobility • Native support for this capability • Wired and Wireless Extensibility • LISP

Canonical Address Format (LCAF) allows for encoding of additional information beyond simply Address-Families.

Extensible

LISP has been actively developed, optimized, and enhanced for the last ten years.

Cisco SD-Access Fabric

Why LISP?

- Optimised resource usage on Edge Nodes:
 - “Pull” only the information needed, like DNS. By comparison BGP pushes all routing information to all Edge Nodes.
- Underlay network is simple and stable:
 - IGP routing from Border Node to Edge Node. Maybe PIM. **No L2, no VLANs, no link bundling, no STP, no MPLS.**
- Unified wired and wireless data plane and policy plane.
 - No wireless concentrator bottleneck = higher throughput.
- Receive future innovations in later SD-Access + IOS XE releases.

Control Plane: Locator/ID Separation Protocol (LISP)

LISP Pub/Sub

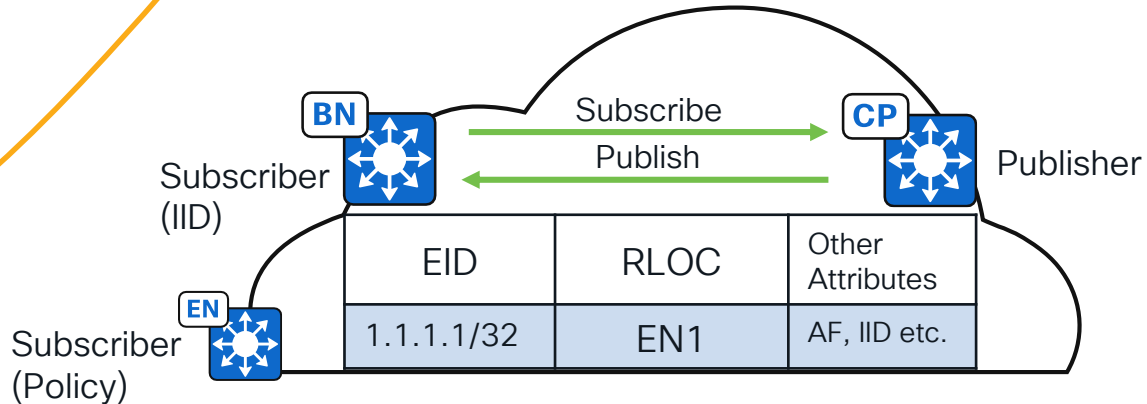
- Released with Catalyst Center 2.2.3.x.
- Reliable and stable.
- Native LISP transport.
- Less Control Plane load.
- Faster convergence.
- Highly extensible.

2022

LISP/BGP

Reliable and stable.
BGP transport.

2017



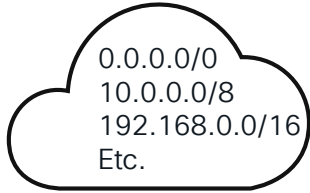
Fabric Operation

LISP Pub/Sub Walkthrough



Fabric Operation

Default ETR Registration

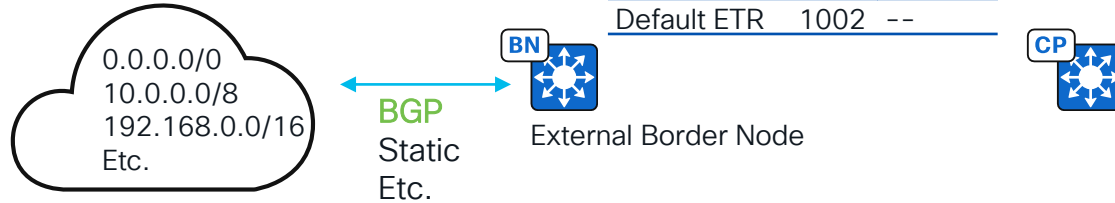


External Border Node



Fabric Operation

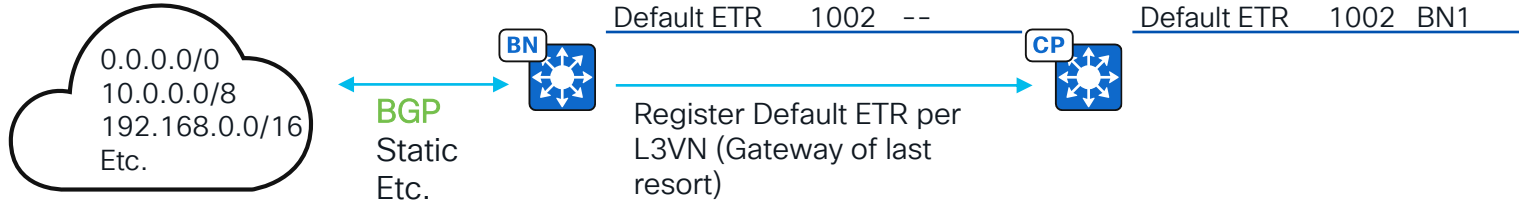
Default ETR Registration



Destination	IID	Next Hop
Default ETR	1001	--
Default ETR	1002	--

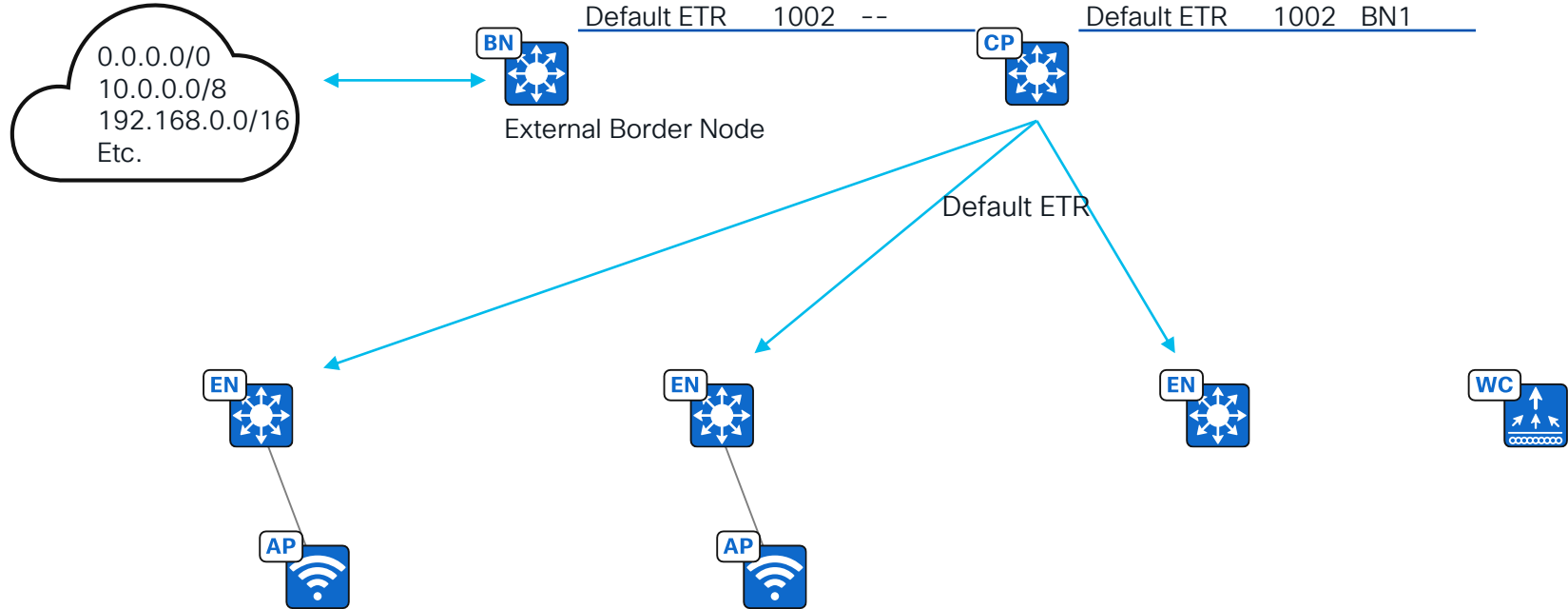
Fabric Operation

Default ETR Registration



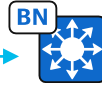
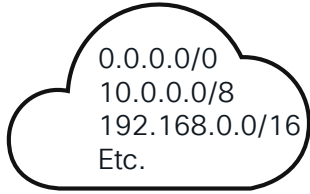
Fabric Operation

Edge Node Bootstrap



Fabric Operation

Edge Node Bootstrap



External Border Node



Destination	IID	Next Hop
Default ETR	1001	--
Default ETR	1002	--

Destination	IID	Next Hop
Default ETR	1001	BN1
Default ETR	1002	BN1



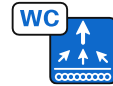
Destination	IID	Next Hop
Default ETR	1001	BN1
Default ETR	1002	BN1



Destination	IID	Next Hop
Default ETR	1001	BN1
Default ETR	1002	BN1

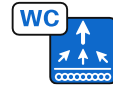
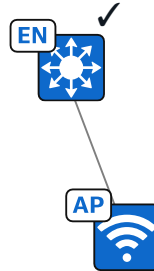
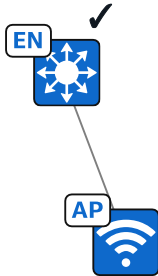
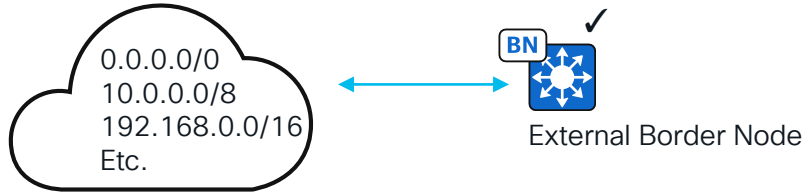


Destination	IID	Next Hop
Default ETR	1001	BN1
Default ETR	1002	BN1



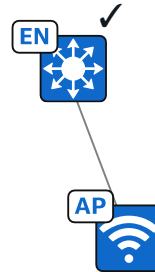
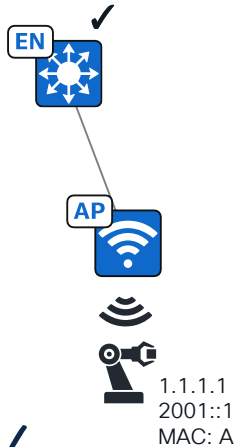
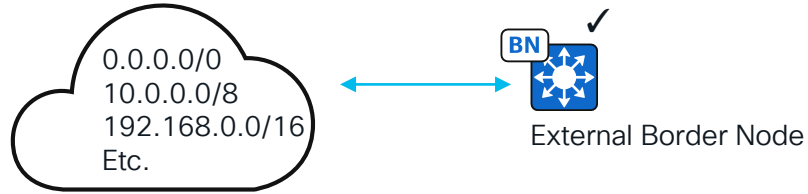
Fabric Operation

Edge Node Bootstrap

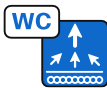
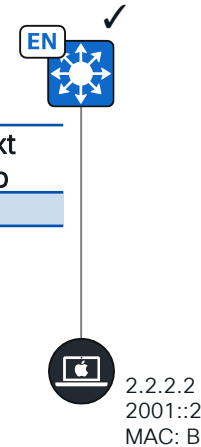


Fabric Operation

Endpoint Registration



Destination	IID	Next Hop
2.2.2.2	1001	--

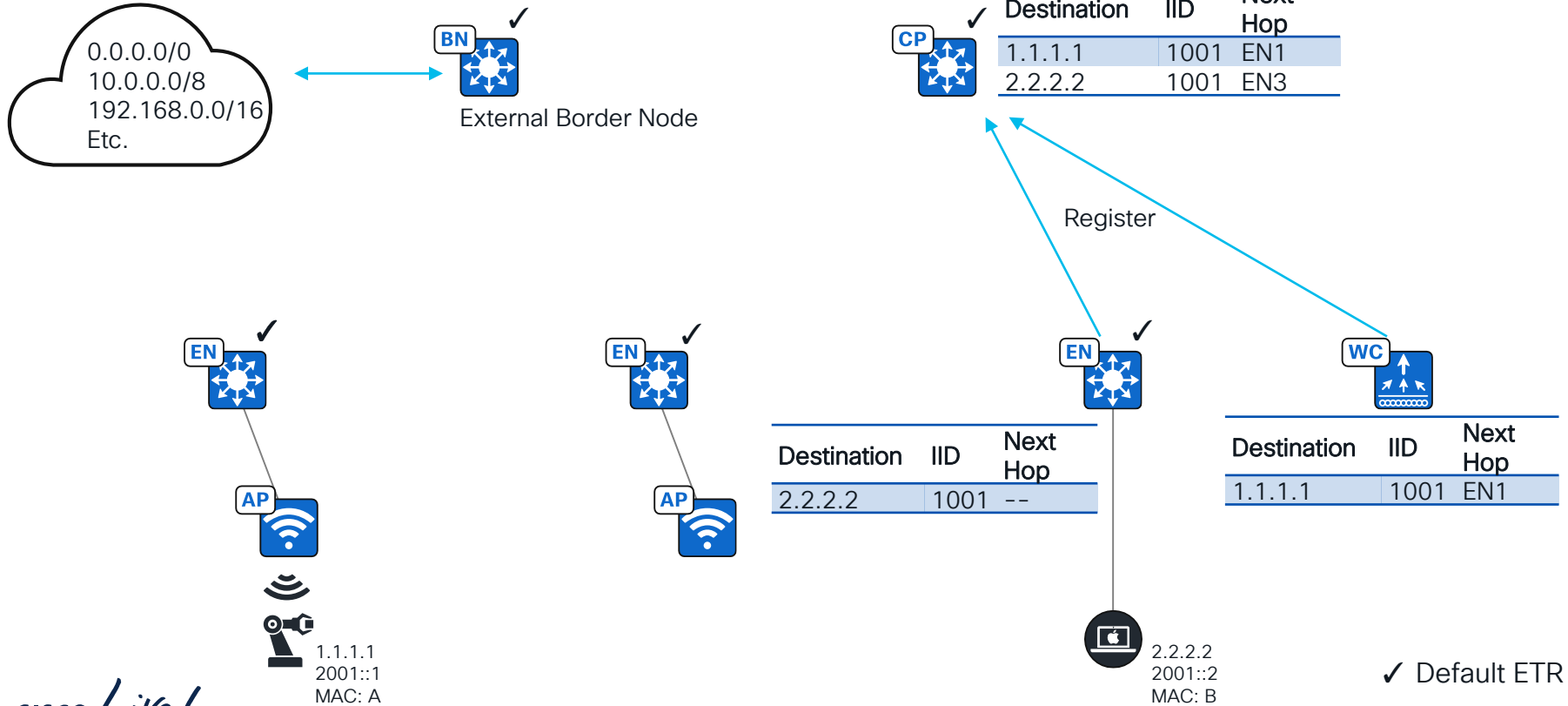


Destination	IID	Next Hop
1.1.1.1	1001	EN1

✓ Default ETR

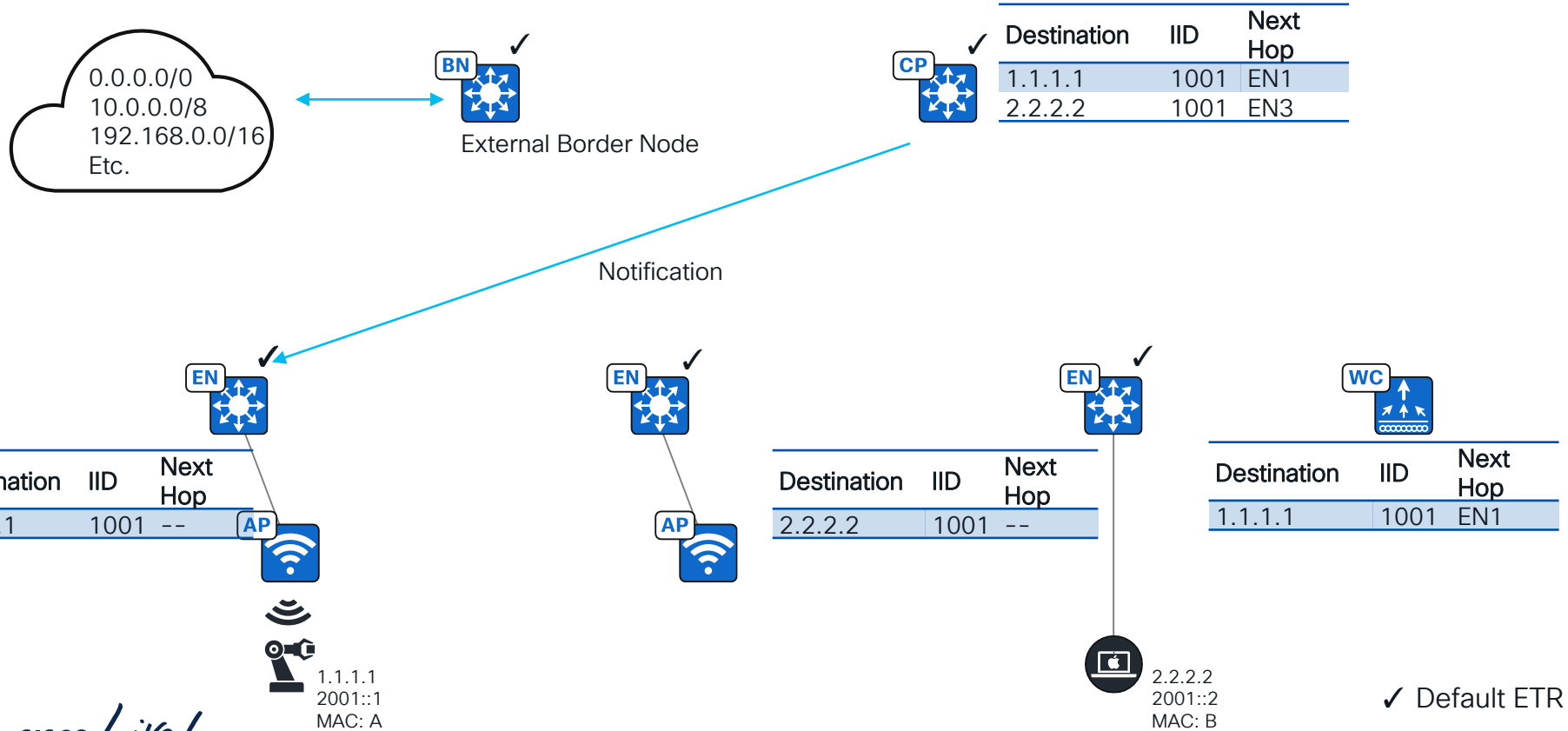
Fabric Operation

Endpoint Registration



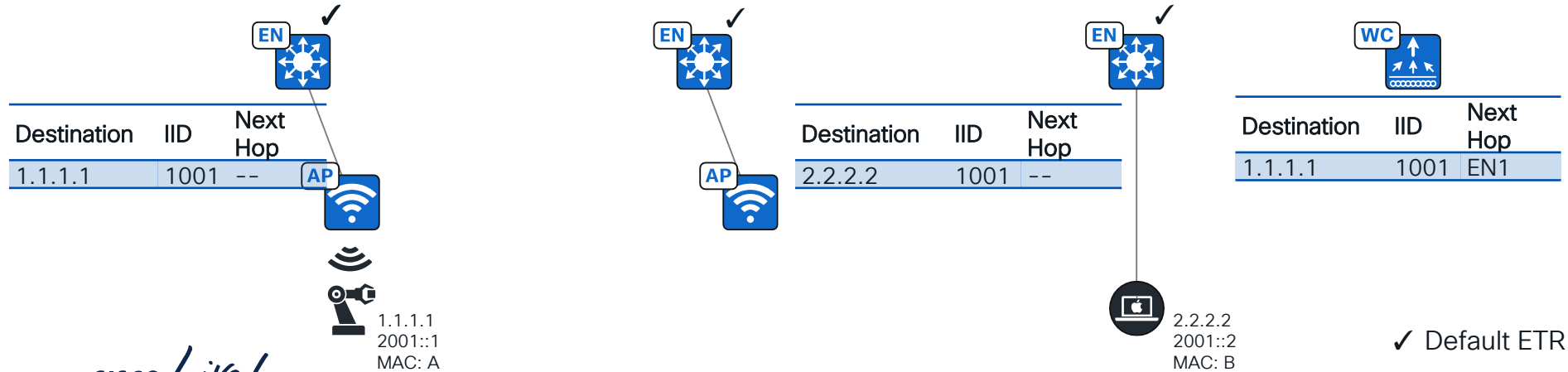
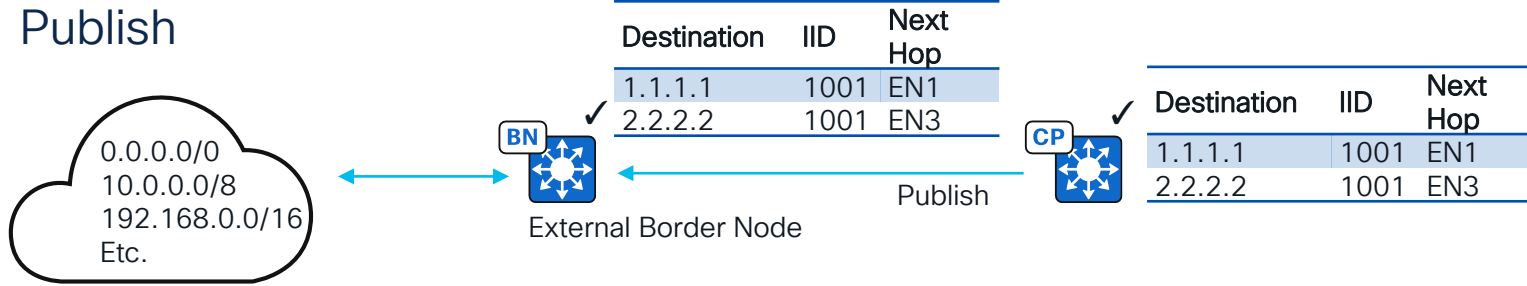
Fabric Operation

Endpoint Registration



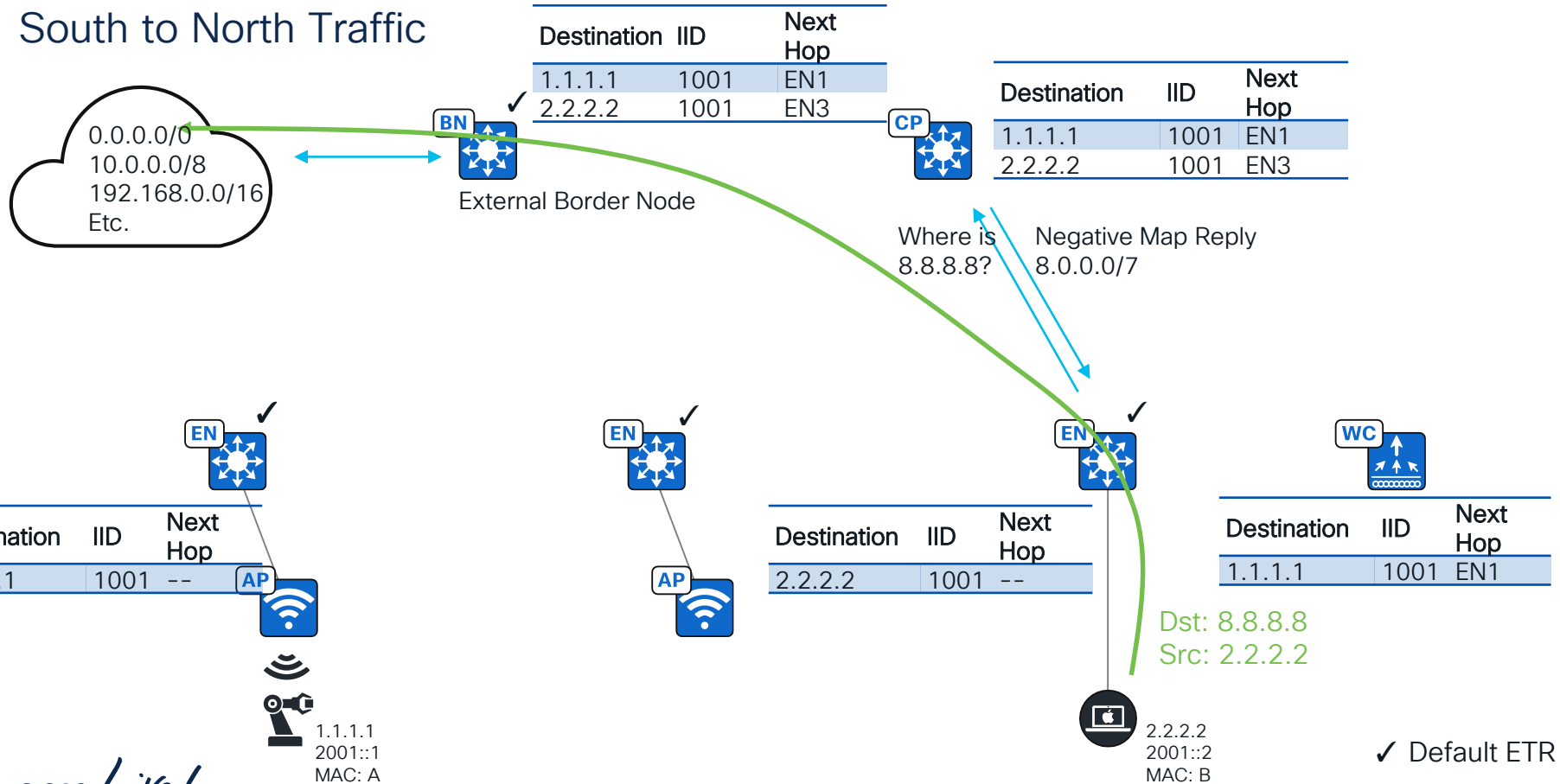
Fabric Operation

Publish



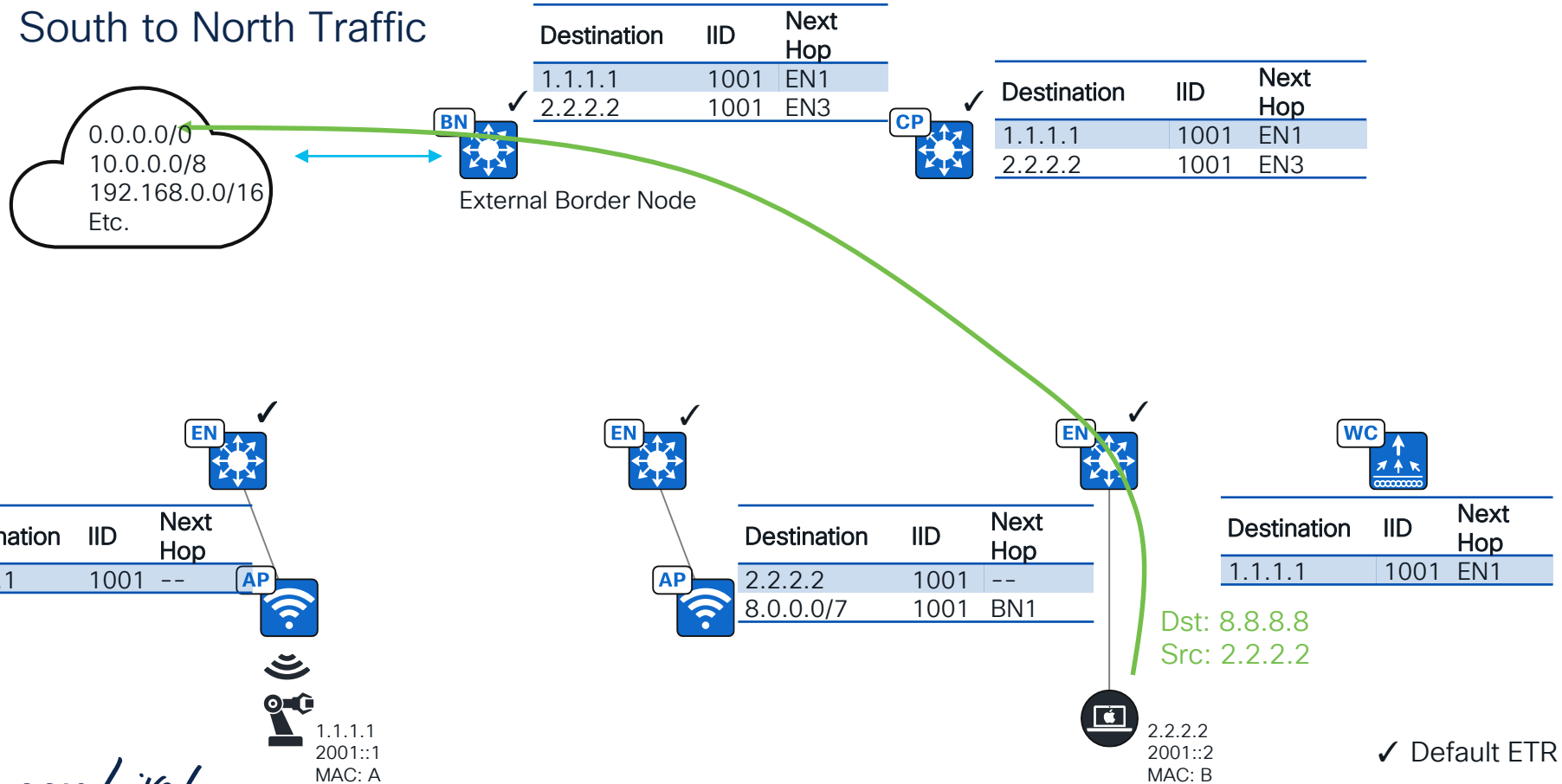
Fabric Operation

South to North Traffic



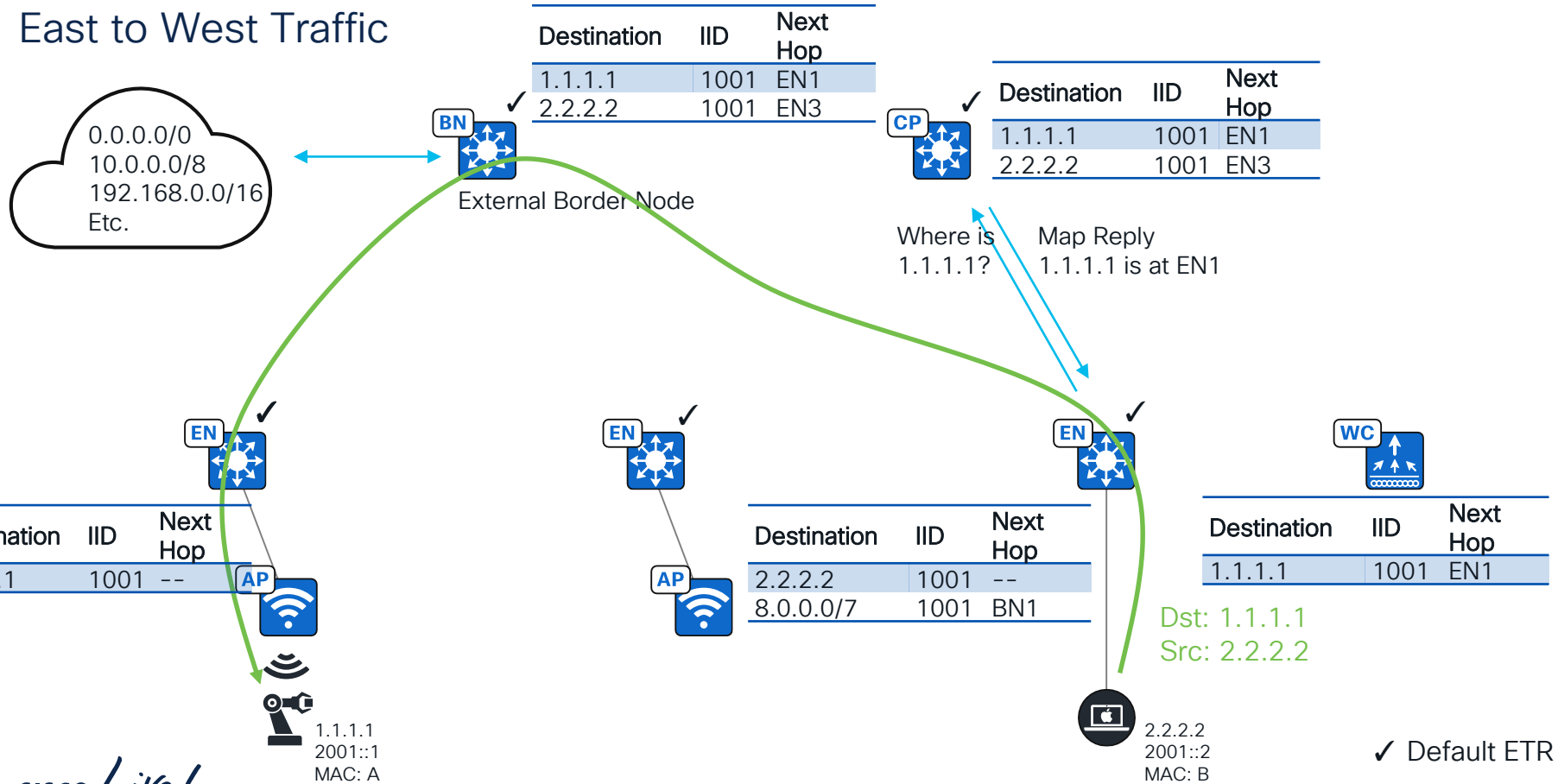
Fabric Operation

South to North Traffic



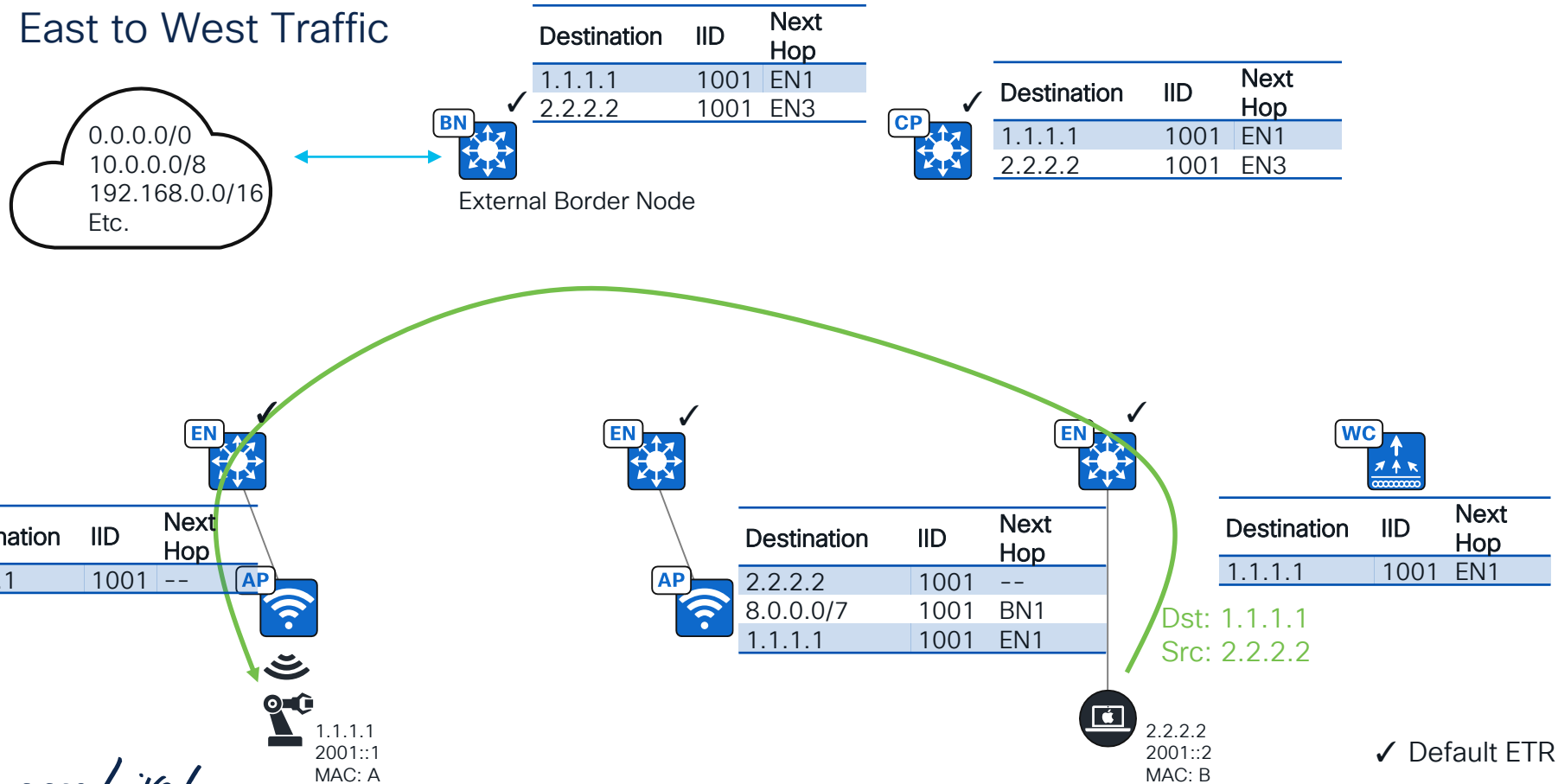
Fabric Operation

East to West Traffic



Fabric Operation

East to West Traffic



LISP Pub/Sub

For Your Information

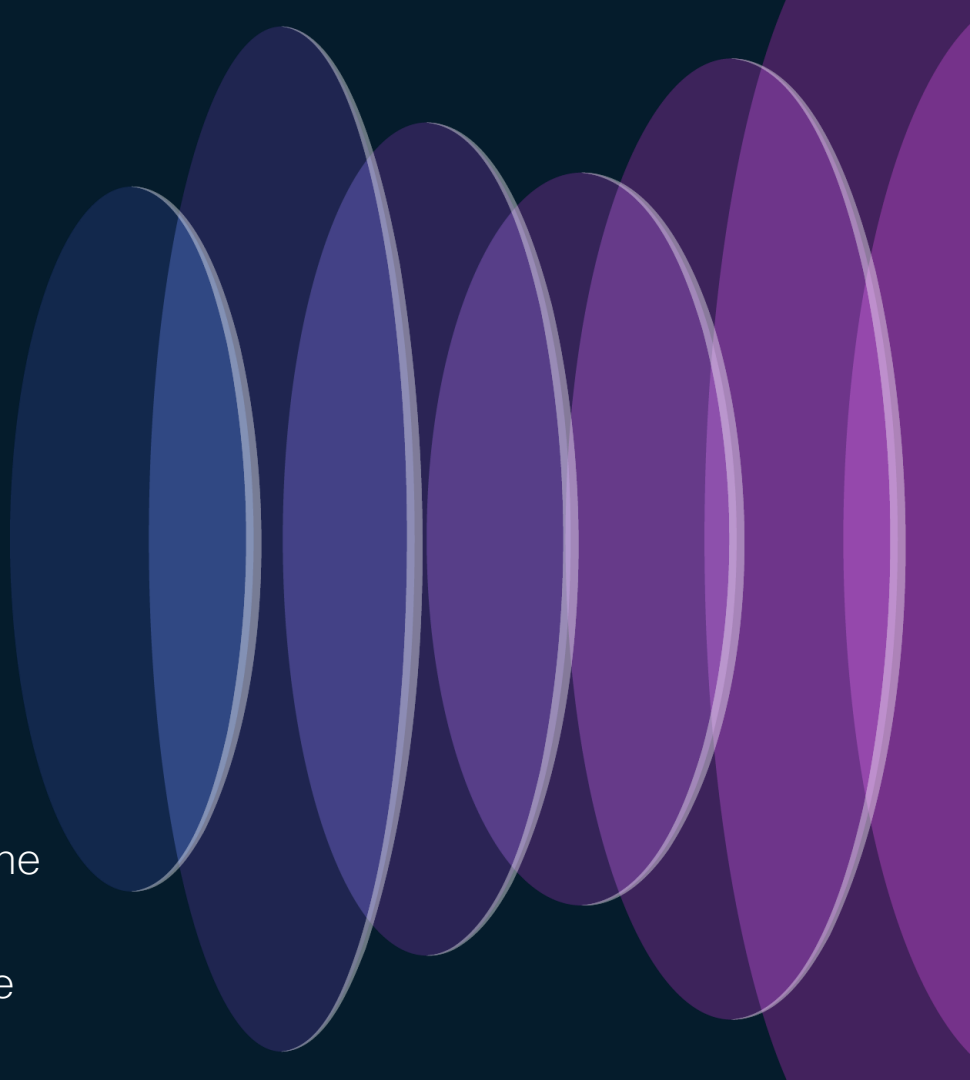


Additional
Information

- No plans to end support for LISP/BGP.
- LISP Pub/Sub is recommended for new deployments.
- In Catalyst Center 2.2.3.x new Fabric Sites can be configured as LISP/BGP or LISP Pub/Sub. Note minimum IOS XE versions.
- LISP/BGP to LISP Pub/Sub migration workflow is under development now.
 - Migrate IP-Based and SD-Access Transit Fabric Sites.
 - ETA CY2024.
- Official release collateral will explain functionality.

Fabric Fundamentals

1. Control Plane
2. Data Plane
3. Policy Plane



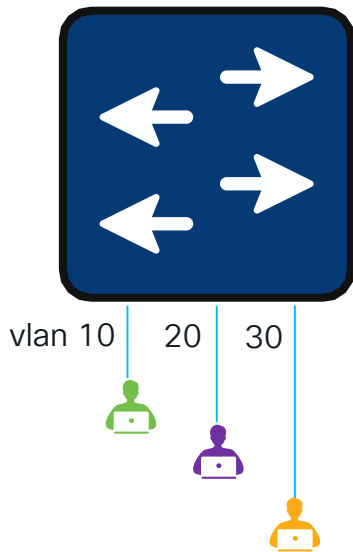


Additional
Information

Cisco SD-Access Fabric

Data Plane: Virtual Extensible Local Area Network (VXLAN)

Traditional Layer 2 Networks Challenges



Spanning Tree Protocol

- Single active path
- Under-utilization
- Broadcast storms

No ECMP

- No load-balancing
- Suboptimal traffic flow

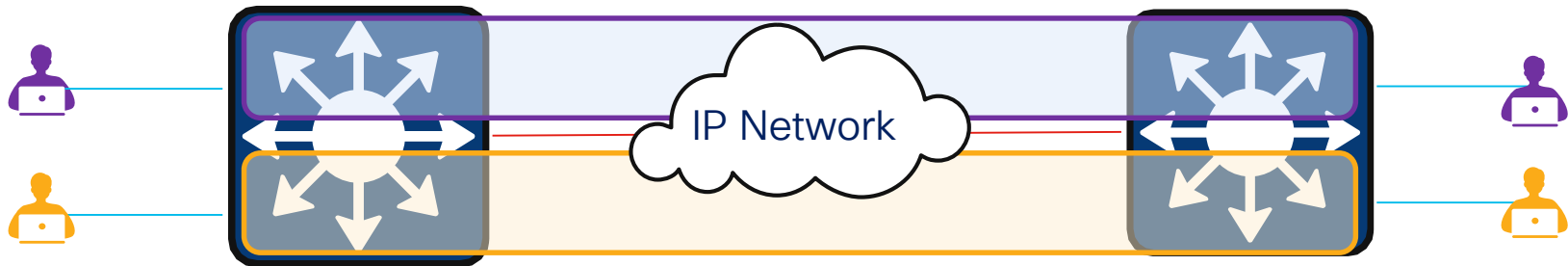
Mobility & Multitenancy

- Restricted to L2 domain
- Geographical limitation
- 12-bit VLAN ID = 4000 VLANs

Cisco SD-Access Fabric

Data Plane: Virtual Extensible Local Area Network (VXLAN)

VXLAN extends Layer 2 and Layer 3 overlay networks over a Layer 3 underlay network



- ✓ Scalability: 16 million unique identifiers.
- ✓ Runs on top of L3, avoids need for STP.
- ✓ L2 traffic tunnelled over an L3 infrastructure.
- ✓ Handles broadcast, multicast, and unknown unicast traffic using multicast instead of flooding.
- ✓ Carries segmentation information.

Cisco SD-Access Fabric

Data Plane: VXLAN

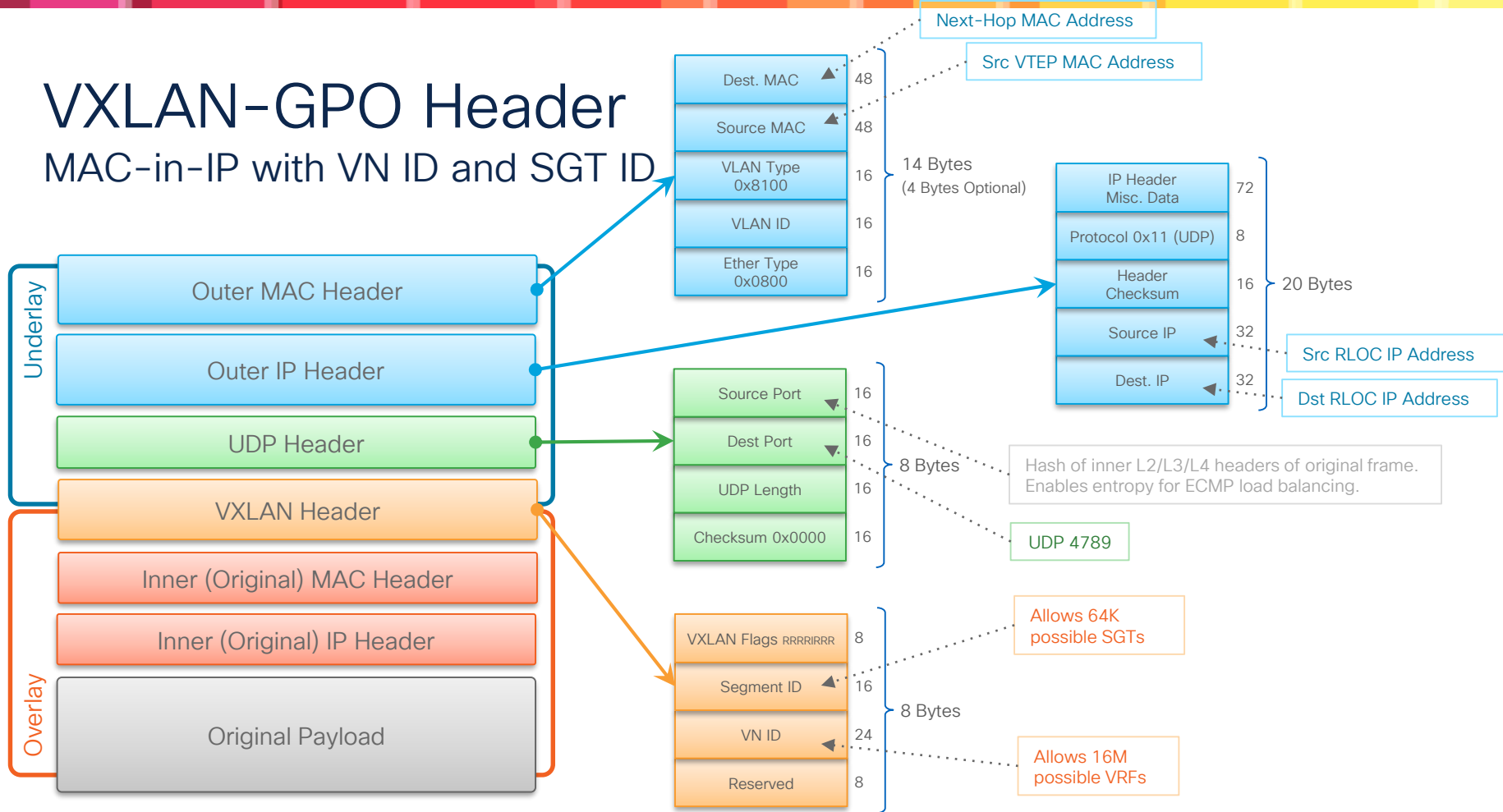
1. Control Plane: LISP
2. Data Plane: VXLAN



Supports L2
& L3 Overlay

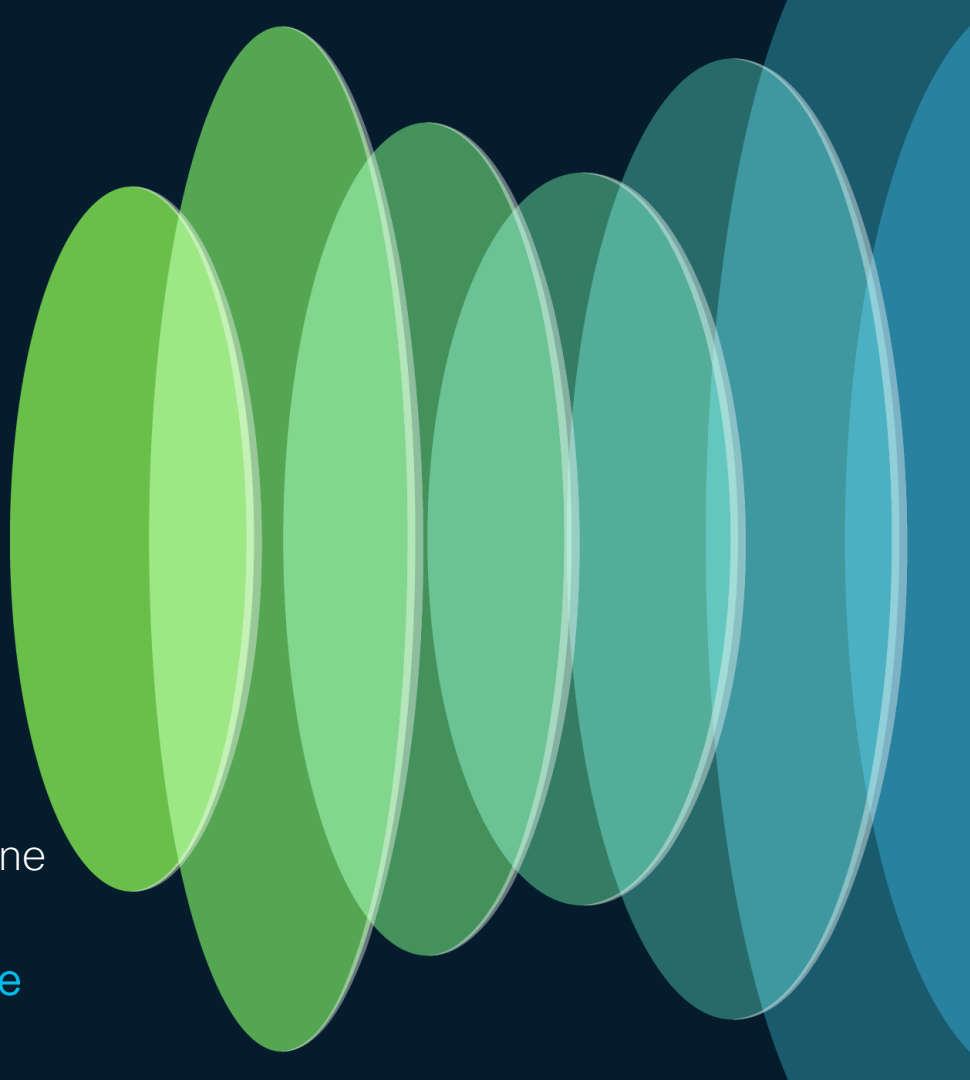
VXLAN-GPO Header

MAC-in-IP with VN ID and SGT ID



Fabric Fundamentals

1. Control Plane
2. Data Plane
3. Policy Plane



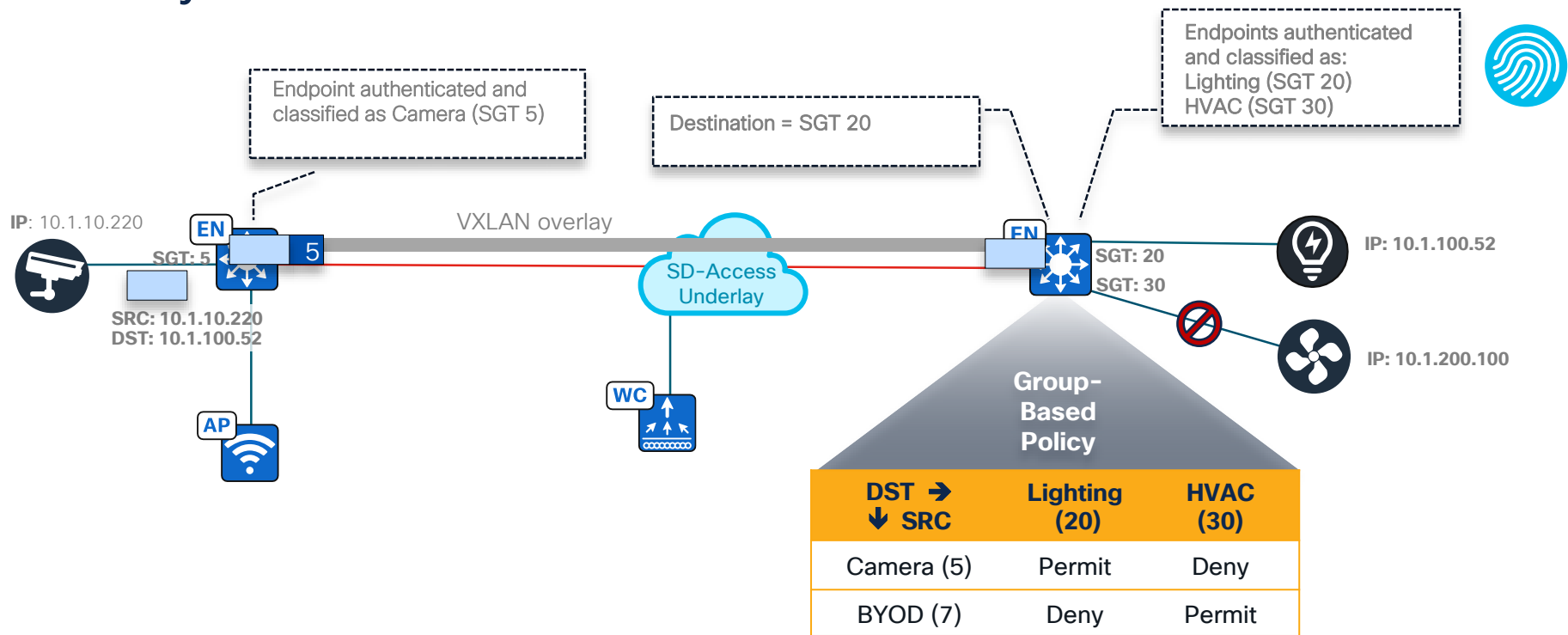
Cisco SD-Access Fabric

Policy Plane: Group-Based Policy

1. Control Plane: LISP
2. Data Plane: VXLAN
3. Policy Plane: Group-Based Policy

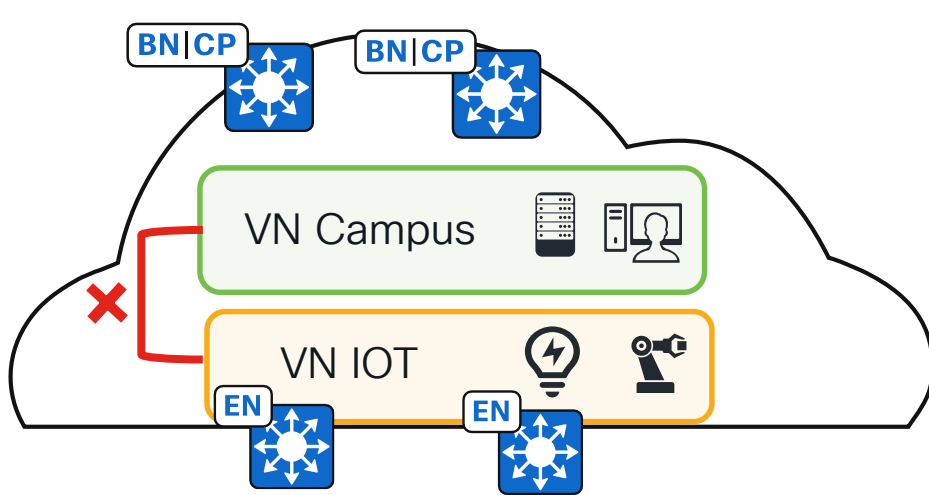


What is Security Group Tag and Group-Based Policy?



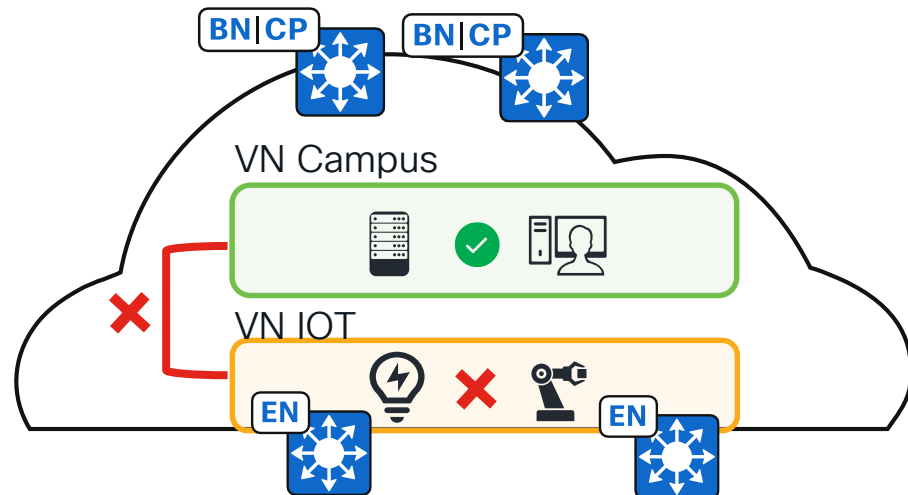
SD-Access Policy

Macro-Segmentation and Micro-segmentation



Virtual Network (VN)

First-level Segmentation ensures **zero communication** between forwarding domains. Ability to consolidate multiple networks into one management plane.

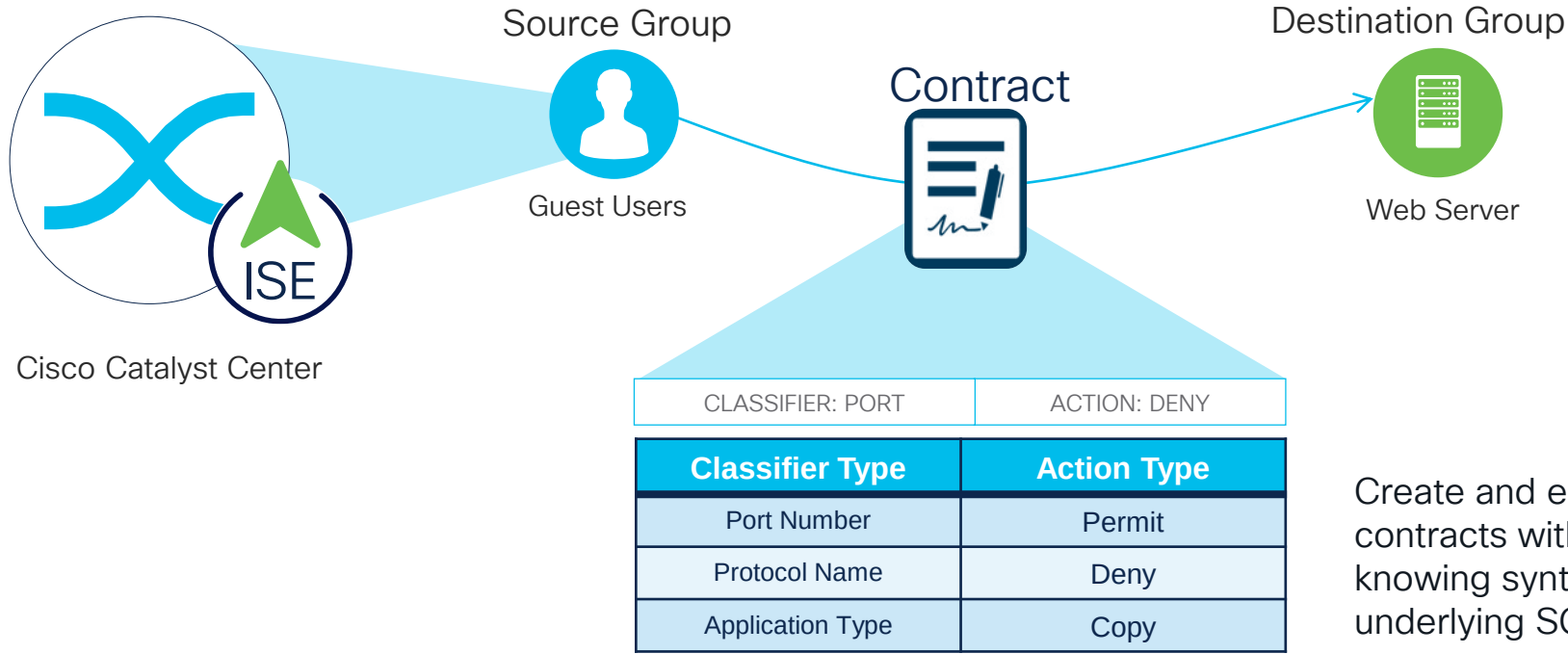


Security Group Tag (SGT)

Second-level Segmentation ensures role-based access control between groups in a VN. Ability to segment the network into lines of business or functional blocks.

SD-Access Policy

Access Control Policies



Create and edit access contracts without knowing syntax for underlying SGACLs.

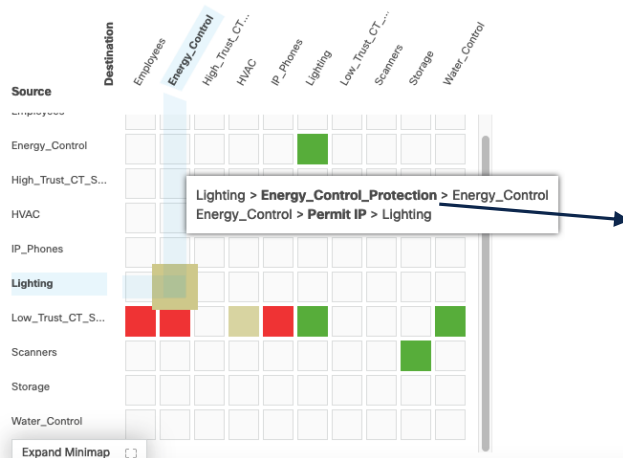
SD-Access Policy

Group-Based Access Control Policy

Policies (11) [Enter full screen](#)

[Filter](#) [Deploy](#) [Refresh](#)

☒ Permit ☐ Deny ☐ Custom ☐ Default



1. Select Source Group(s)
2. Select Destination Group(s)
3. Select Access Contract(s)

Access Contract

Name Description

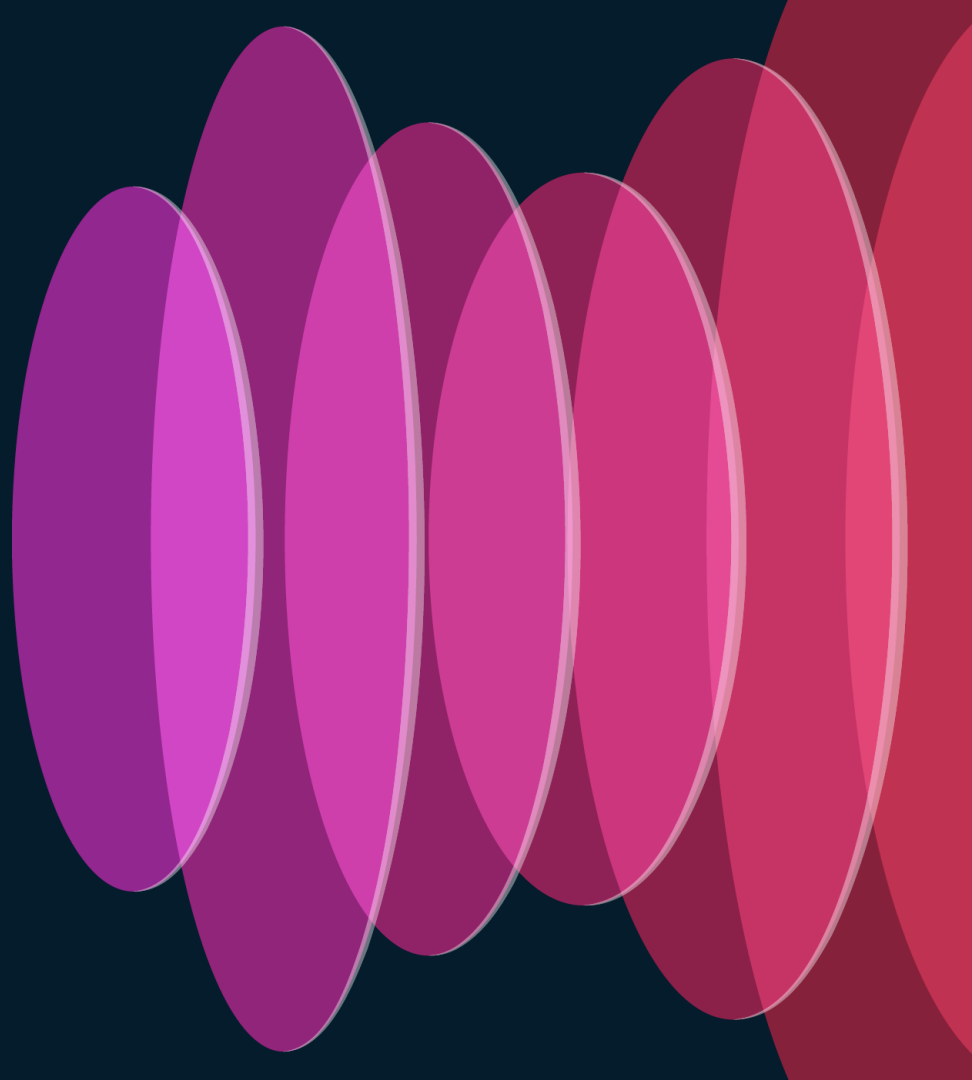
Energy_Control_Protection

CONTRACT CONTENT (1)

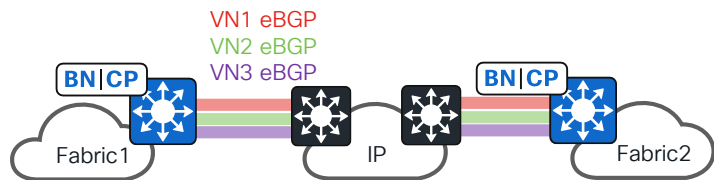
#	Action	Application	Transport Protocol	Source / Destination	Port	Logging
1	Permit	https	TCP/UDP	Destination	443/443	OFF

Default Action Permit Logging OFF

Multiple Fabrics



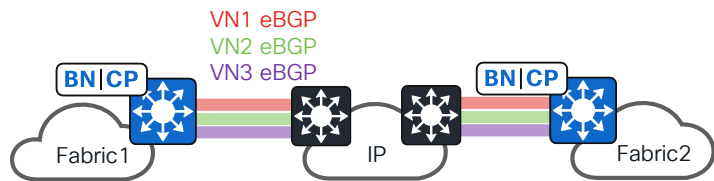
Transits for VN and SGT Preservation



IP-Based Transit

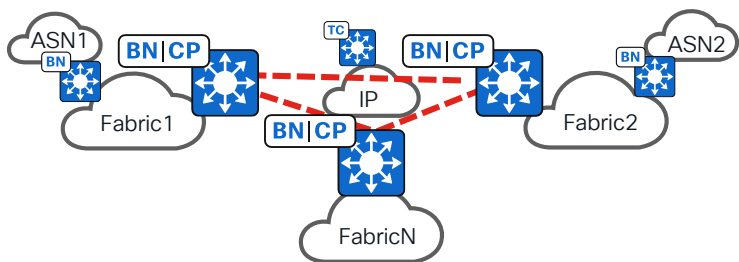
- Per-Layer-3-Virtual-Network eBGP peering to external routing domain, or LISP Extranet Provider VN eBGP peering to external routing domain.
- SGT propagation outside of fabric requires suitable hardware and software.

Transits for VN and SGT Preservation



IP-Based Transit

- Per-Layer-3-Virtual-Network eBGP peering to external routing domain, or LISP Extranet Provider VN eBGP peering to external routing domain.
- SGT propagation outside of fabric requires suitable hardware and software.

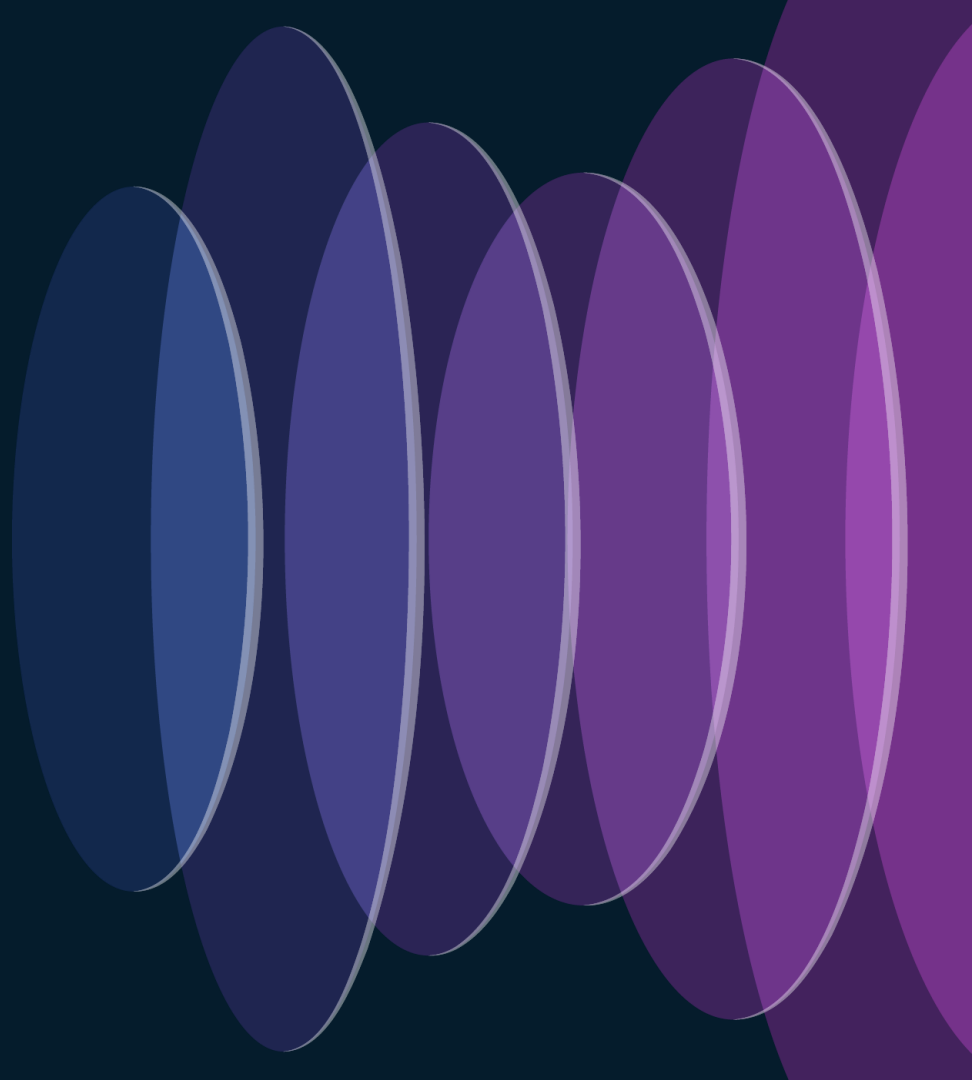


SD-Access Transit

- SD-Access LISP/VXLAN between Fabric Sites.
- Preserves Layer 3 Virtual Networks and SGT.
- Fabric as a transit between external routing domains.



Conclusion



Cisco's SD-Access LISP provides a secure, flexible, and automated way to meet the security and operational challenges faced by an ever-changing environment.



Global Partner Solution Advisors

NEW - Fully Virtualized, SD-Access Secure Campus Lab

Virtualized SD-Access Lab

- Fully Customizable Topology with virtualized 9kv's and 8kv's
- Access on dCloud or build on your existing Data Center
- Fraction of the cost
- GPSA mentored lab buildout support available!



Virtual SD-Access
Lab on dCloud



GPSA Sales
Connect Page



CTF at Cisco Live
Check out Secure
Campus Section

CTF Mission

- Experience the SD-Access Virtual Lab at Capture the Flag in The World of Solutions
- Use Cases – Fabric Sites and Virtual Network Provisioning, Fusion Automation, Extranet, Micro Segmentation, and more!

Contact

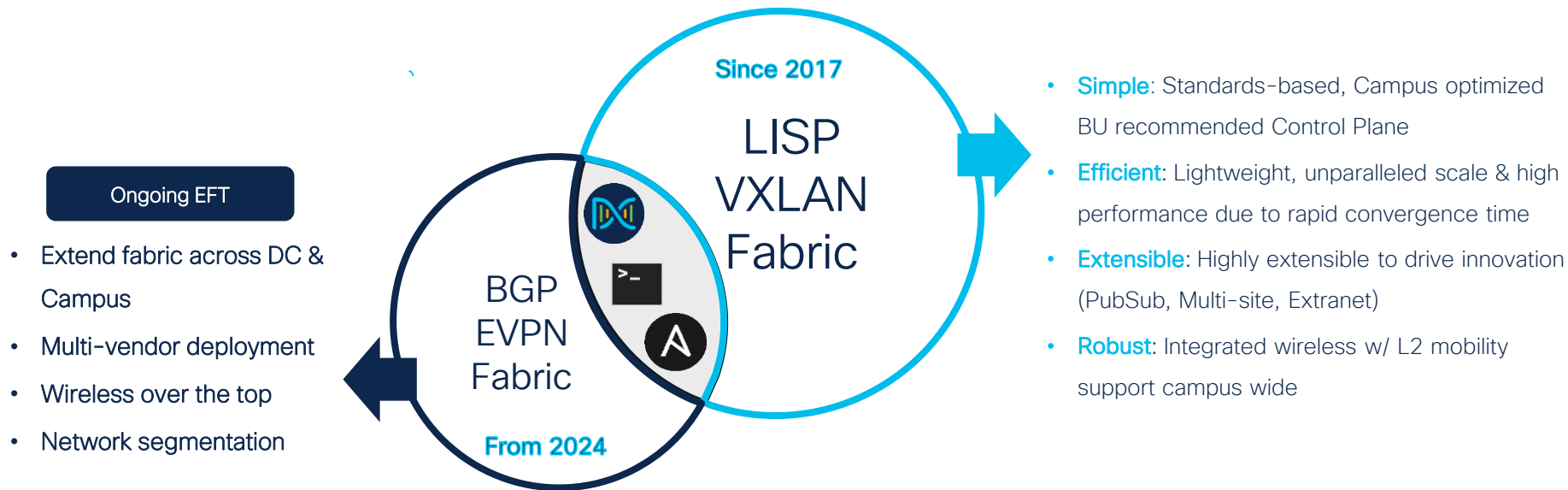
- GPSA is your source for **no-cost**, partner enablement and practice building!
- Visit the Global Partner Experience booth (4227) across from Capture the Flag, for more information.

cisco Live!



Effortlessly Deploy Your Fabric of Choice

LISP Fabric is the **leading** choice for Enterprise customers!



One Infrastructure | Single Data Plane | Consistent Zero-Trust Experience

SD-Access LISP Industry Leading Campus Architecture



Deployments
4050+



Momentum
40%

YoY growth in customers



Key use case

70%

Wireless

+ 66%

API (YoY)



Usage

24K+

Sites

1.8M+

Devices



Top verticals: Government, Finance,
Professional services, and Manufacturing

Adopted by 31% of U.S. Fortune 100
Companies

EMEA: 52%

Americas 29%

APJC 19%

SD-Access LISP Customer Success

Healthcare



Education + Energy

Yale



Manufacturing



SCALE

5300 devices
15K+endpoints

6200 devices
10K+endpoints

REQUIREMENTS

Zero-Trust Network Access
HIPAA Compliance

6500 devices
66K+endpoints

5300 devices
57K+endpoints

Segmentation at scale
Automated operations
APIs for Automation & Tool Integration

4500 devices
10K+endpoints

16k devices
98K+endpoints

Secure, Highly available network
Hi performance scalable WI-FI

Segmentation at Scale | Unified Wired/Wireless Policy | IT/OT Integration Experience

CISCO *Live!*

BRKENS 1801, BRKENS 1802, CIUG-1003

Speaking at this Cisco Live

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Catalyst Leadership in Enterprise Networks

A Platform based Approach

Catalyst Center and Meraki Dashboard

28M Network Devices Managed

↑ 50% Y/Y 19M APs | 6M Switches | 2.5M Routers | 830M Clients

13M
Devices on
Catalyst Center



15.3M
Devices on
Meraki Dashboard



Catalyst 9000 Family



100,000+ Customers, Millions of Switches

“Catalyst 9K continues to be the fastest ramping product in the company's history”
- Chuck Robbins, CEO Cisco Systems

cisco Live!

Secure Networking

Common
Policy

Secure Equipment
Access

SD-Access
(LISP & EVPN)

High-speed
Encryption

Digital Experience

Campus Automation

AI Endpoint
Analytics

Digital Experience
ThousandEyes

AI Ops & Assurance

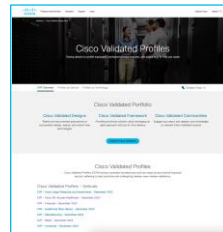
Operational Simplicity

Cloud Managed
Catalyst

Infrastructure
as a Code

S3 & CloudWatch
Integration

Visibility, Control
& Rollback



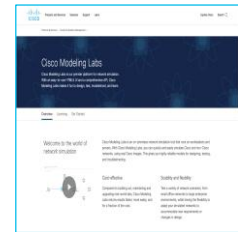
Cisco Validated Profiles
(CVP)



Industry
Validated Reports

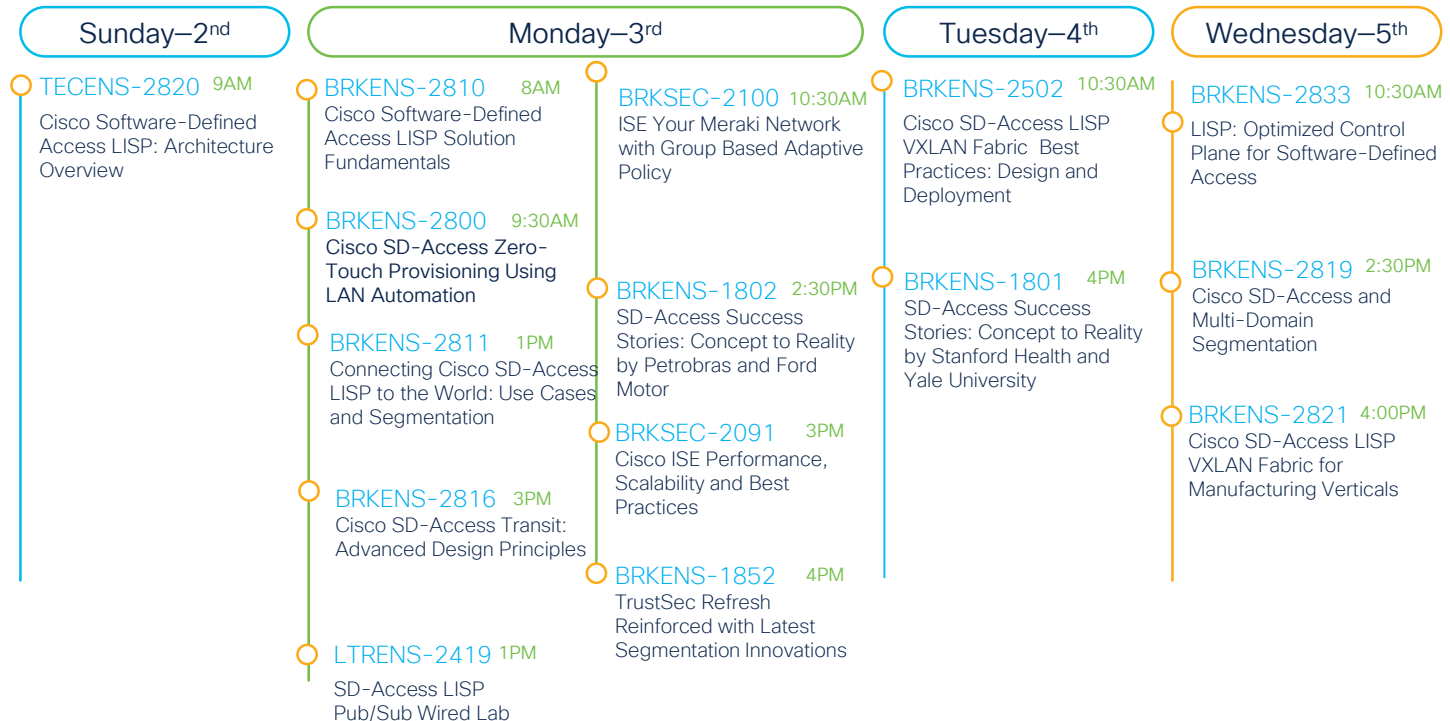


Industry
Certifications



Cisco Modeling
Labs

Cisco Live US SD-Access and ISE Learning Map



● BU-led sessions

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Cisco SD-Access Collaterals



[Cisco Software-Defined Access for Industry Verticals](#)



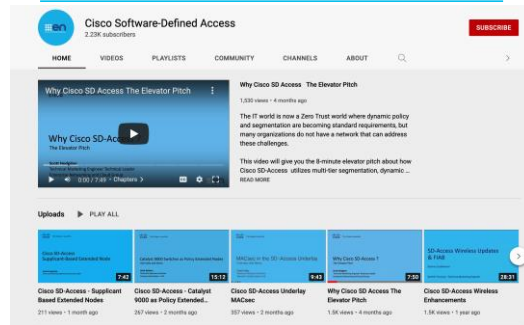
[Cisco Software-Defined Access](#) [Enabling intent-based networking](#)



[Cisco Solution Validated Profiles \(CVPs\)](#)

- [Cisco Large Enterprise and Government Profile](#)
- [Healthcare Vertical](#)
- [Financial Vertical](#)
- [Healthcare Vertical](#)
- [Manufacturing Vertical](#)
- [Retail Vertical](#)
- [University Vertical](#)

[Cisco SD-Access YouTube Link](#)



[Multiple Cisco Catalyst Center to ISE](#)

[Cisco SD-Access Design Tool](#)

[EN&C Validated Designs](#)

[The Latest SD-Access Guides](#)

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- Book your one-on-one Meet the Engineer meeting
- Attend the interactive education with DevNet, Capture the Flag, and Walk-in Labs
- Visit the On-Demand Library for more sessions at www.CiscoLive.com/on-demand



The bridge to possible

Thank you

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